Canonical Digamma Polynomials and the Fifth Harmonic Bridge

Laurent coefficients, polygamma primitives,
and finite reconstruction of even-tail moments

Research continuation prepared for the ProveIt project

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Abstract

We continue the exact-identity programme in the ProveIt manuscript and its recent incoming reports. The fifth coordinate finite part of the digamma function is evaluated in terms of the linear Laurent coefficients of strict multiple zeta functions of depths two, three, and four, together with one explicitly specified holomorphic mixed double-zeta derivative. All finite endpoint constants are proved by finite harmonic algebra and a normalized Mittag–Leffler expansion. A Gamma quotient provides a meromorphic realization of the classical Roman harmonic Bell polynomials. Its coefficients have exactly prescribed strict-harmonic simple-pole residues and satisfy an ordinary integral identity isolating one height-one Laurent coefficient at every depth. We prove uniqueness, up to constants, of the scalar Gamma–Appell normalization that cancels all nondecaying residue defects, and give explicit polygamma differential corrections, primitives, and polylogarithmic integral kernels. Independently, an exponential-polynomial argument proves that an explicit finite prefix reconstructs every centered moment of two even harmonic tails; equal orders admit an exact determinant and the first q moments. A concrete orders-$(2,6)$ consequence is an absolutely convergent rational polygamma sum. Prior results are attributed, the remaining stronger conjectures are separated from the proved statements, and a narrow published Stieltjes sign-description error is corrected. Exact symbolic replays and independent numerical diagnostics accompany the proofs.

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1 Scope, exact targets, and conventions

The starting point is the manuscript directory and the incoming archive directory of the ProveIt repository, inspected at the commit printed on the title page [1]. A repository-relative claim of progress is different from a claim of priority in the mathematical literature. This article makes the former precise. Classical Gamma quotients, harmonic Bell polynomials, height-one coefficient series, and low-weight multiple-zeta reductions are explicitly attributed. The proofs developed here organize these tools into a fifth-power bridge and a finite moment reconstruction theorem. The bounded literature review is not a proof that no equivalent formulation exists elsewhere.

Two especially relevant incoming reports are *Quartic Tails and Complex Resolvents* and *Barnes Resolvents and Higher Harmonic Identities* [2, 3]. They already establish the quartic digamma bridge, the all-depth reciprocal-Gamma singular Laurent coefficients, and finite formulas for two-even-tail moments. Their stated further questions, rather than results already proved in their bodies, determine our targets.

Source target	Result and precise boundary
Quartic report, Q1: fifth digamma power	Theorem 5.1 gives the complete fifth-power finite part, its mixed coordinate, and all endpoint constants.
Quartic report, Q2: canonical polynomial	The scalar residue normalization is proved for every degree, uniquely up to an additive constant. An exact differential normalization is also proved. The stronger claim that scalar polynomials remove every lower-depth arithmetic coordinate remains open.
Quartic report, Q8: initial-moment generation	Theorem 7.1 proves an explicit finite-prefix bound and rational reconstruction. For equal orders the first q moments suffice. The same first-q assertion for all unequal orders remains open.
Manuscript S_6, revised S_8	No proof is asserted here. Existing numerical residual certificates retain their original evidentiary status.

1.1 Harmonic and spectral notation

Put $H_n^{(r)} = \sum_{k=1}^{n} k^{-r}$ and $H_n = H_n^{(1)}$, with every empty sum zero. For strict indices, use

$$H_n(s_1, \dots, s_d) = \sum_{n \geq k_1 > \cdots > k_d \geq 1} k_1^{-s_1} \cdots k_d^{-s_d}, \tag{1.1}$$

$$Z_d(s_1, \dots, s_d) = \sum_{k_1 > \cdots > k_d \geq 1} k_1^{-s_1} \cdots k_d^{-s_d} \tag{1.2}$$

in their convergence regions, followed by analytic continuation where specified. The elementary harmonic polynomials are

$$e_j(n) = H_n(\{1\}^j), \qquad e_0(n) = 1, \qquad \sum_{j \geq 0} e_j(n) t^j = \prod_{k=1}^{n} \left(1 + \frac{t}{k}\right). \tag{1.3}$$

They must be distinguished from the complete, weakly ordered harmonic sums, whose generator has factors $(1 - t/k)^{-1}$.

The ordinary Stieltjes convention is

$$\zeta(1+u) = \frac{1}{u} + \sum_{j \geq 0} \frac{(-1)^j \gamma_j}{j!} u^j, \qquad \gamma = \gamma_0. \tag{1.4}$$

At depth d, define literal regular Laurent coefficients by

$$\lambda_{d,j} = [u^j] Z_d(1 + u, \{1\}^{d-1}), \qquad \lambda_d = \lambda_{d,1}. \tag{1.5}$$

The coefficient brackets impose neither a factorial nor an alternating sign. In particular

$$\lambda_1 = -\gamma_1, \qquad \lambda_2 = \eta, \quad \lambda_3 = \theta, \quad \lambda_4 = \xi.$$

The ordered germ has a pole of order d, not a simple pole for $d > 1$. It is therefore unsafe to identify it with a Hurwitz-zeta germ merely by calling both sets of coefficients generalized Stieltjes constants.

We use the reciprocal-Gamma coefficients

$$\frac{1}{\Gamma(1+t)} = \sum_{j\geq 0} c_j t^j, \quad c_0 = 1, \quad c_1 = \gamma, \quad c_2 = \frac{\gamma^2 - \zeta(2)}{2}, \quad c_3 = \frac{\gamma^3 - 3\gamma\zeta(2) + 2\zeta(3)}{6}. \tag{1.6}$$

They are obtained recursively by differentiating the standard log-Gamma expansion [10]:

$$nc_n = \gamma c_{n-1} + \sum_{r=2}^{n} (-1)^{r+1} \zeta(r) c_{n-r}. \tag{1.7}$$

1.2 A finite part is a specified constant coefficient

For a function meromorphic at zero and smooth on $(0, 1]$, throughout this article

$$\mathrm{FP} \int_0^1 f(x)\, dx = \mathrm{CT}_{\varepsilon\downarrow 0} \int_\varepsilon^1 f(x)\, dx. \tag{1.8}$$

The expansion uses powers of the coordinate ε itself and $\log \varepsilon$. There is no silently changed cutoff, rescaled coordinate, or substituted spectral regularization. If $f(x) = \sum_{k=-M}^{\infty} a_k x^k$ near zero, this definition can be computed without divergent numerical cancellation as

$$\int_0^1 \left(f(x) - \sum_{k=-M}^{-1} a_k x^k \right) dx + \sum_{k=-M}^{-2} \frac{a_k}{k+1}. \tag{1.9}$$

The contribution from a_{-1}/x is zero. For a meromorphic F,

$$\mathrm{FP} \int_0^1 F'(x)\, dx = F(1) - [x^0]F(x). \tag{1.10}$$

This rule follows directly by differentiating the Laurent series; it will account for every endpoint term in the fifth-power calculation.

All ordinary integrals below are identified as such. All matrix ranks in the moment reconstruction are formal rational ranks. Neither finite-part calculations nor rational elimination require unproved arithmetic independence of zeta or Stieltjes constants.

2 A normalized Mellin formula for the height-one germs

This section gives a self-contained normalization of the coefficient data used later. The reciprocal-Gamma singular coefficients and height-one series are established objects in the literature [4, 5] and in the incoming Barnes report [3]. We record the precise integral form so that the fifth power and the ordinary Gamma integral share a proved convention.

For $v > 0$, put

$$B(v) = -\log(1 - e^{-v}), \qquad V_d(v) = B(v)^d - \mathbf{1}_{(0,1)}(v)(-\log v)^d. \tag{2.1}$$

Near zero, $B(v) = -\log v + O(v)$, and consequently

$$V_d(v) = O\big(v(1 + |\log v|^{d-1})\big).$$

At infinity it decays exponentially. Thus $\int_0^\infty v^{u-1} V_d(v)\, dv$ is holomorphic for $\mathrm{Re}\, u > -1$; each derivative on a compact subset is obtained by inserting the corresponding power of $\log v$.

Proposition 2.1 (Mellin normalization)**.** *For $d \geq 1$, initially for $\operatorname{Re} u > 0$ and then meromorphically for $\operatorname{Re} u > -1$,*

$$\boxed{\Gamma(1+u)Z_d(1+u,\{1\}^{d-1}) = \frac{1}{u^d} + \frac{u}{d!}\int_0^\infty v^{u-1}V_d(v)\,dv.} \tag{2.2}$$

In particular the complete principal part and the constant term are

$$Z_d(1+u,\{1\}^{d-1}) = \sum_{j=0}^{d} c_j u^{j-d} + \lambda_d u + O(u^2), \tag{2.3}$$

where

$$\lambda_d = c_{d+1} + \frac{1}{d!}\int_0^\infty \frac{V_d(v)}{v}\,dv. \tag{2.4}$$

Proof. The generating identity

$$\operatorname{Li}_{\{1\}^{d-1}}(y) = \frac{(-\log(1-y))^{d-1}}{(d-1)!}$$

follows either from strict harmonic generating functions or by repeated integration of $1/(1-y)$. Termwise Mellin integration gives

$$\Gamma(1+u)Z_d(1+u,\{1\}^{d-1}) = \frac{1}{(d-1)!}\int_0^\infty \frac{v^u B(v)^{d-1}}{e^v-1}\,dv.$$

For $\operatorname{Re} u > 0$ the integral is absolutely convergent. Since $B'(v) = -1/(e^v-1)$, integration by parts, with both boundary terms zero, turns it into

$$\frac{u}{d!}\int_0^\infty v^{u-1}B(v)^d\,dv.$$

The subtracted comparison integral is exactly $\int_0^1 v^{u-1}(-\log v)^d\,dv = d!/u^{d+1}$. The estimates preceding the proposition justify continuation and all coefficient extractions. Multiplication by the Taylor series (1.6) proves the remaining formulas. □

Corollary 2.2 (Every regular coefficient and spectral derivative)**.** *Let*

$$I_{d,r} = \int_0^\infty \frac{(\log v)^r V_d(v)}{v}\,dv \qquad (d \geq 1,\ r \geq 0).$$

For every $j \geq 1$,

$$\lambda_{d,j} = c_{d+j} + \frac{1}{d!}\sum_{r=0}^{j-1} c_{j-1-r}\frac{I_{d,r}}{r!}. \tag{2.5}$$

If $\Lambda_d(u) = \Gamma(1+u)Z_d(1+u,\{1\}^{d-1}) - u^{-d}$, then

$$\Lambda_d^{(j)}(0) = \frac{j}{d!}I_{d,j-1} \qquad (j \geq 1). \tag{2.6}$$

These are ordinary convergent logarithmic Mellin integrals. At depth one, $\lambda_{1,j} = (-1)^j\gamma_j/j!$ recovers the usual Stieltjes derivative convention.

Proof. Expand the holomorphic integral in (2.2) into its Taylor series and multiply by $\sum c_j u^j$. The estimates are uniform on every closed disc $|u| \leq \rho < 1$, so the operations are analytic coefficient extractions rather than formal regularizations. □

Formula (2.5) is useful for reproducibility: coefficients of different depths can be evaluated in one fixed scheme, with no matching of independently chosen subtraction constants. It is also a direct route to higher spectral derivatives whenever a new digamma identity produces them.

3 The Gamma quotient and its exact harmonic residues

The generating function in this section is classical at integer arguments: it generates the Roman harmonic numbers, or complete harmonic star sums, discussed by Sesma [7, Eq. (49)]. We use its meromorphic continuation in the harmonic cutoff to produce a convenient exact differential normalization of the digamma powers. The all-depth ordinary integral below is proved directly; its coefficient-series data are related to Young's established formulas, and its first case is Coffey's integral for γ_1 [8, Eq. (1.19)].

Definition 3.1. For x away from the nonpositive integers, define

$$G(x,t) = \frac{\Gamma(1-t)\Gamma(x)}{\Gamma(x-t)} = \sum_{m\geq 0} \mathcal{C}_m(x)\frac{t^m}{m!}. \tag{3.1}$$

Coefficient extraction is at $t = 0$; the resulting functions of x are extended meromorphically.

Put

$$h_1(x) = \psi(x) + \gamma, \qquad h_r(x) = \zeta(r) - \zeta(r,x) = \zeta(r) + \frac{(-1)^{r+1}}{(r-1)!}\psi^{(r-1)}(x) \quad (r \geq 2). \tag{3.2}$$

The log-Gamma Taylor expansion yields

$$\log G(x,t) = \sum_{r\geq 1} h_r(x)\frac{t^r}{r}. \tag{3.3}$$

Consequently $\mathcal{C}_m = Y_m(h_1, 1!h_2, \ldots, (m-1)!h_m)$, where Y_m is the complete exponential Bell polynomial. In particular

$$\mathcal{C}_0 = 1, \qquad \mathcal{C}_1 = h_1, \qquad \mathcal{C}_2 = h_1^2 + h_2, \tag{3.4}$$

$$\mathcal{C}_3 = h_1^3 + 3h_1h_2 + 2h_3, \tag{3.5}$$

$$\mathcal{C}_4 = h_1^4 + 6h_1^2h_2 + 3h_2^2 + 8h_1h_3 + 6h_4, \tag{3.6}$$

$$\mathcal{C}_5 = h_1^5 + 10h_1^3h_2 + 15h_1h_2^2 + 20h_1^2h_3 + 20h_2h_3 + 30h_1h_4 + 24h_5. \tag{3.7}$$

At every degree a finite recurrence suffices:

$$\mathcal{C}_{m+1} = \sum_{r=0}^{m} \binom{m}{r} r!h_{r+1}\mathcal{C}_{m-r}. \tag{3.8}$$

At a positive integer n the generator is $\prod_{k=1}^{n-1}(1-t/k)^{-1}$, so $\mathcal{C}_m(n)/m!$ is the weakly ordered harmonic sum of m unit indices.

Theorem 3.2 (Exact poles and removable values)**.** *For $m \geq 2$, $\mathcal{C}_m$ has only simple poles, with*

$$\operatorname{Res}_{x=-n} \mathcal{C}_m(x) = -m!e_{m-1}(n), \qquad n \geq 0. \tag{3.9}$$

The points $0, -1, \ldots, -(m-2)$ are removable. Its first actual pole is $-(m-1)$ and has residue $-m$. At the removable points,

$$\boxed{\mathcal{C}_m(-j) = -m!\sum_{k=0}^{j} e_k(j)\zeta(m-k), \qquad 0 \leq j \leq m-2.} \tag{3.10}$$

In particular $\mathcal{C}_m(0) = -m!\zeta(m)$, and $\mathcal{C}_m(1) = 0$.

Proof. The reciprocal Gamma factor is entire, so the Gamma quotient has at most simple poles in x. The Gamma recurrence gives exactly

$$\operatorname{Res}_{x=-n} G(x,t) = \frac{(-1)^n \Gamma(1-t)}{n!\Gamma(-n-t)} = -t \prod_{k=1}^{n} \left(1 + \frac{t}{k}\right).$$

Taking coefficients proves the residue assertion. Since $e_{m-1}(n)$ vanishes precisely for $n < m-1$ and is positive thereafter, it also proves the removability and the first-pole statements.

For the values, set $x = -j + \varepsilon$ and $E_j(t) = \prod_{k=1}^{j}(1 + t/k)$. Expansion of both Gamma factors gives the constant coefficient in ε as

$$-tE_j(t)\big(H_j - \gamma - \psi(-j-t)\big).$$

By the digamma recurrence this equals

$$tE_j(t)(\gamma + \psi(1-t)) - tE_j(t)H_j + E_j(t) + tE_j(t)\sum_{k=1}^{j} \frac{1}{k+t}. \tag{3.11}$$

All terms after the first are polynomials of degree at most $j+1$. For the last term this follows by cancelling the corresponding factor of E_j. Finally $\gamma + \psi(1-t) = -\sum_{r\geq 1} \zeta(r+1)t^r$. At degree $m \geq j+2$ only this series contributes, proving (3.10). Since the pole coefficient at that degree is zero, its finite Laurent value is the ordinary removable value. The identity at 1 follows from $G(1,t) = 1$. □

Theorem 3.3 (A normally convergent partial-fraction expansion)**.** *For $m \geq 2$, away from its actual poles,*

$$\boxed{\mathcal{C}_m(z) = -m!\zeta(m) - m! \sum_{n\geq 1} e_{m-1}(n) \left(\frac{1}{n+z} - \frac{1}{n}\right).} \tag{3.12}$$

The series converges normally on every compact subset of that domain. Terms whose coefficient $e_{m-1}(n)$ is zero are omitted, including at a removable point where their displayed denominator vanishes.

Proof. Since $e_{m-1}(n) = O((1+\log n)^{m-1})$, the subtracted series converges normally. To determine the entire remainder, integrate $z\mathcal{C}_m(w)/(w(w-z))$ on $|w| = N + 1/2$. The right-half-plane digamma asymptotics and their derivatives, combined with reflection on the left, give $\mathcal{C}_m(w) = O((1+\log N)^m)$ uniformly on these circles. The distance to the nonpositive integral poles stays bounded below. In the reflection formula the cotangent and its fixed derivatives are bounded there, while the remaining polygammas have their usual sectorial bounds [10]. The integral tends to zero like $O((1+\log N)^m/N)$.

Its residues at $w = z$ and $w = 0$ are $\mathcal{C}_m(z)$ and $-\mathcal{C}_m(0)$. Each pole $w = -n$ contributes $-m!e_{m-1}(n)z/(n(n+z))$. The residue theorem and Theorem 3.2 give (3.12). Thus matching the principal parts alone is not being used to discard an arbitrary entire function. □

3.1 An ordinary integral isolates one strict-depth coefficient

Theorem 3.4 (The all-depth ordinary-integral bridge)**.** *For every integer $m \geq 2$,*

$$\boxed{\int_0^1 \mathcal{C}_m(x)\,dx = m!(c_m - \lambda_{m-1}).} \tag{3.13}$$

The integrand extends continuously to both endpoints. There is no finite-part operation in this identity.

Proof. Use the opposite-sign parameter $F(x,t) = G(x,-t) = \Gamma(1+t)\Gamma(x)/\Gamma(x+t)$ and define

$$\Phi(t) = \int_0^1 \{F(x,t) - 1 + t(\psi(x)+\gamma)\}\, dx.$$

Its origin residue is $t - t = 0$. More precisely, the subtracted integrand is jointly holomorphic in t near zero and continuous in $x \in [0,1]$; this follows by subtracting t/x from the Gamma quotient. Uniform coefficient extraction on $|t| \le \rho < 1$ gives

$$\Phi(t) = \sum_{m\ge 2} \frac{(-1)^m t^m}{m!} \int_0^1 \mathcal{C}_m(x)\, dx. \tag{3.14}$$

For $\operatorname{Re} t > 0$, Euler's beta integral and the standard digamma integral give, after combining their integrands,

$$\Phi(t) = t\int_0^\infty \left\{ \frac{(1-e^{-v})^t - 1}{v} + \frac{e^{-v}}{1-e^{-v}} \right\} dv - 1.$$

The combination is integrable at both ends. For example the original integrand before integration in x is $e^{-xv}((1-e^{-v})^{t-1} - (1-e^{-v})^{-1}) + e^{-v}/(1-e^{-v})$; its joint integral is absolutely convergent when $\operatorname{Re} t > 0$, which justifies Fubini. Integrate the second term from a lower cutoff δ and use $\int_\delta^\infty e^{-v}/(1-e^{-v})\, dv = -\log(1-e^{-\delta})$. Comparing with $\int_\delta^1 (v^t - 1)\, dv/v$ yields

$$\Phi(t) = t\int_0^\infty \frac{(1-e^{-v})^t - 1 - \mathbf{1}_{(0,1)}(v)(v^t-1)}{v}\, dv. \tag{3.15}$$

This subtracted integral is holomorphic for $\operatorname{Re} t > -1$ locally about zero. Near zero, $(1-e^{-v})^t - v^t = O(v^{\operatorname{Re} t+1})$ uniformly on compact parameter sets; at infinity the numerator decays exponentially. Thus the equality continues to $|t| < 1$.

At degree $m \ge 2$, its right side has coefficient

$$\frac{(-1)^{m-1}}{(m-1)!} \int_0^\infty \frac{V_{m-1}(v)}{v}\, dv.$$

Comparison with (3.14) gives $\int_0^1 \mathcal{C}_m = -mI_{m-1,0}$. Equation (2.4) proves the claimed formula. □

The first two instances are

$$\int_0^1 \{(\psi+\gamma)^2 + \zeta(2) - \psi'\}\, dx = \gamma^2 - \zeta(2) + 2\gamma_1, \tag{3.16}$$

$$\int_0^1 \{(\psi+\gamma)^3 + 3(\psi+\gamma)(\zeta(2)-\psi') + 2\zeta(3) + \psi''\}\, dx = 6(c_3 - \eta). \tag{3.17}$$

The apparent singularities in each complete integrand cancel; its individual displayed terms should not be separately integrated as ordinary improper integrals. The first formula rearranges exactly to Coffey's known identity cited above.

3.2 The known Cauchy-series coordinates recovered from the same quotient

An additional beta representation is useful both for interpretation and for stable numerical evaluation:

$$\mathcal{C}_m(x) = (x-1)\int_0^1 (-\log y)^m (1-y)^{x-2}\, dy, \qquad m \ge 1, \quad \operatorname{Re} x > 1-m. \tag{3.18}$$

For $\operatorname{Re} x > 1$ it follows by differentiating $G(x,t) = (x-1)B(1-t, x-1)$. The integral on the right then continues it to the stated half-plane: near $y = 1$ the exponent is $x + m - 2$. This also independently displays the location of the first pole.

Define positive rational coefficients

$$a_n = \frac{1}{(n-1)!}\int_0^1 (r)_n\, dr, \qquad a_1 = \frac{1}{2}, \quad a_2 = \frac{5}{6}, \quad a_3 = \frac{9}{8}, \tag{3.19}$$

where $(r)_n = r(r+1)\cdots(r+n-1)$ is a rising factorial. If $L = -\log(1-y)$, their generating kernel is

$$K(y) = \sum_{n\geq 1} a_n y^{n-1} = \frac{L-y}{(1-y)^2 L^2}. \tag{3.20}$$

Indeed integrate $(1-y)^{-r} = \sum (r)_n y^n/n!$ over $r \in [0,1]$ and differentiate the resulting $y/((1-y)L)$. Integrating (3.18) in x, for $m \geq 2$, gives

$$\int_0^1 \mathcal{C}_m(x)\, dx = -\int_0^1 (-\log y)^m K(y)\, dy.$$

The integrals are absolutely convergent; alternatively Tonelli applies to their nonnegative negatives. The elementary Mellin integral of y^{n-1} therefore yields

$$\boxed{\lambda_d = c_{d+1} + \sum_{n\geq 1} \frac{a_n}{n^{d+2}}, \qquad d \geq 1.} \tag{3.21}$$

This coefficient series is an equivalent form of the established Cauchy/Bernoulli-second-kind series in Young's work [4, 5]; it is included as an independent normalization check and a connection with the meromorphic differential realization, not as a new coefficient formula.

4 A scalar Gamma–Appell normalization of digamma residues

The differential polynomials generated by a normalized Gamma quotient have exactly the strict harmonic residues required for a single-depth Laurent coefficient. There is also a canonical scalar polynomial in the digamma function that has the same residues up to a decaying correction. The distinction is useful: scalar powers are the original finite-part integrands, whereas differential polynomials remove all unwanted pole orders at once.

For $n \geq 0$, let

$$e_j(n) = \sum_{1\leq i_1<\cdots<i_j\leq n} \frac{1}{i_1\cdots i_j}, \qquad e_0(n) = 1, \qquad e_j(n) = 0 \quad (j > n). \tag{4.1}$$

Thus $e_j(n)$ is a strict finite harmonic sum, and

$$E_n(t) := \sum_{j\geq 0} e_j(n) t^j = \prod_{k=1}^{n} \left(1 + \frac{t}{k}\right) = \frac{\Gamma(n+1+t)}{\Gamma(n+1)\Gamma(1+t)}. \tag{4.2}$$

Define the monic polynomials

$$P_m(Y) = m![t^m]\Gamma(1-t)e^{tY}, \qquad m \geq 0. \tag{4.3}$$

They satisfy $P_0 = 1$ and $P_m'(Y) = mP_{m-1}(Y)$, so they form an Appell sequence. For example,

$$\begin{aligned} P_1(Y) &= Y + \gamma, \\ P_2(Y) &= (Y+\gamma)^2 + \zeta(2), \\ P_3(Y) &= (Y+\gamma)^3 + 3\zeta(2)(Y+\gamma) + 2\zeta(3). \end{aligned} \tag{4.4}$$

Every coefficient assertion below is at a fixed finite order; no uniformity as $m \to \infty$ is required. The generating function also gives an explicit finite recurrence:

$$P_{m+1}(Y) = (Y+\gamma)P_m(Y) + \sum_{r=1}^{m} \binom{m}{r} r!\zeta(r+1)P_{m-r}(Y). \tag{4.5}$$

It follows by differentiating with respect to t and using $-\psi(1-t) = \gamma + \sum_{r\geq 1}\zeta(r+1)t^r$.

Lemma 4.1 (The universal cotangent residue)**.** *For every $t \in \mathbb{C}$,*

$$\operatorname*{Res}_{v=0} e^{-\pi t \cot(\pi v)} = -\frac{\sin(\pi t)}{\pi}. \tag{4.6}$$

Here the residue is the coefficient of v^{-1}; the isolated singularity may be essential.

Proof. Put $w = \pi \cot(\pi v)$. Near $v = 0$, this is an invertible change between the local coordinate v and the local coordinate $1/w$ at infinity, since $w = v^{-1} + O(v)$. Moreover,

$$\frac{dw}{dv} = -(w^2 + \pi^2), \qquad e^{-\pi t \cot(\pi v)}\, dv = -\frac{e^{-tw}}{w^2 + \pi^2}\, dw. \tag{4.7}$$

Residues of differentials are invariant under this local coordinate change. The two finite residues of the differential on the right are $-e^{-i\pi t}/(2i\pi)$ at $w = i\pi$ and $e^{i\pi t}/(2i\pi)$ at $w = -i\pi$. Their sum is $\sin(\pi t)/\pi$. Applying the residue theorem on a circle containing these two points, or equivalently taking the residue at infinity, gives its negative, as asserted. This argument applies to every complex t; a pole rather than an essential singularity at infinity is not needed for that contour calculation. □

Theorem 4.2 (Scalar residue normalization and uniqueness)**.** *For $m \geq 1$ and $n \geq 0$, set*

$$r_{m,n} = \operatorname*{Res}_{x=-n} P_m(\psi(x)), \qquad \delta_{m,n} = r_{m,n} + m! e_{m-1}(n). \tag{4.8}$$

Then $\delta_{1,n} = 0$, and for each fixed $m \geq 2$,

$$\delta_{m,n} = O_m\left(\frac{(1 + \log n)^{m-2}}{n}\right) \qquad (n \to \infty). \tag{4.9}$$

In particular, the difference in (4.8) tends to zero.

Conversely, a monic polynomial $Q \in \mathbb{C}[Y]$ of degree m satisfies

$$\operatorname*{Res}_{x=-n} Q(\psi(x)) + m! e_{m-1}(n) \longrightarrow 0 \tag{4.10}$$

if and only if $Q - P_m$ is constant. Consequently $P_m(Y) - P_m(0)$ is the unique such monic polynomial with zero constant term.

Proof. Fix a circle $|v| = \rho$ with $0 < \rho < 1/2$. The reflection formula and the uniform large-argument expansion of the digamma function give

$$\psi(-n + v) = \psi(n + 1 - v) - \pi \cot(\pi v) = L_n + K(v) + \eta_n(v), \tag{4.11}$$

where $L_n = \log n$, $K(v) = -\pi \cot(\pi v)$, and η_n is holomorphic on $|v| \leq \rho$ with $\sup |\eta_n| = O(n^{-1})$. Only $n \geq 1$ is needed for this asymptotic argument.

Consider, coefficientwise near $t = 0$,

$$R_n(t) = \operatorname*{Res}_{v=0} e^{tK(v) + t\eta_n(v)}. \tag{4.12}$$

Integration on the fixed circle makes $R_n(t)$ an entire function of t. By Lemma 4.1,

$$R_n(t) = -\frac{\sin(\pi t)}{\pi} + \Delta_n(t), \qquad \Delta_n(t) = \operatorname*{Res}_{v=0} e^{tK(v)}\big(e^{t\eta_n(v)} - 1\big). \tag{4.13}$$

Each fixed Taylor coefficient of Δ_n is $O(n^{-1})$: this follows directly by bounding its contour integral on $|v| = \rho$ for t on a fixed small circle. More precisely, $\Delta_n(0) = 0$ and $\Delta_n'(0) = \operatorname{Res}_{v=0} \eta_n(v) = 0$. Thus Δ_n starts with t^2, a fact that gives the exponent $m - 2$ in (4.9).

Taking coefficients in (4.3) and using (4.11)–(4.13) therefore yields, for $m \geq 2$,

$$\begin{aligned} r_{m,n} &= -m![t^m]\Gamma(1-t)e^{tL_n}\frac{\sin(\pi t)}{\pi} + O_m\left(\frac{(1+L_n)^{m-2}}{n}\right) \\ &= -m![t^{m-1}]\frac{e^{tL_n}}{\Gamma(1+t)} + O_m\left(\frac{(1+L_n)^{m-2}}{n}\right). \end{aligned} \tag{4.14}$$

The last equality is Euler's reflection identity $\Gamma(1-t)\sin(\pi t)/\pi = t/\Gamma(1+t)$.

For comparison, uniformly for t in a sufficiently small fixed disk,

$$\frac{\Gamma(n+1+t)}{\Gamma(n+1)} = e^{tL_n}e^{u_n(t)}, \qquad u_n(0) = 0, \qquad [t^j]u_n(t) = O_j(n^{-1}) \quad (j \geq 1). \tag{4.15}$$

For example, this follows by integrating $\psi(n+1+s) - \log n = O(n^{-1})$ from $s = 0$ to $s = t$. Consequently the correction $e^{u_n(t)} - 1$ starts with t and has each fixed coefficient $O(n^{-1})$. By (4.2), for $d \geq 1$,

$$e_d(n) = [t^d]\frac{e^{tL_n}}{\Gamma(1+t)} + O_d\left(\frac{(1+L_n)^{d-1}}{n}\right). \tag{4.16}$$

Equations (4.14) and (4.16), with $d = m-1$, prove (4.9). For $m = 1$ the assertion is immediate from $P_1(\psi) = \psi + \gamma$, whose residue at every nonpositive integer is -1.

It remains to prove necessity and uniqueness. Let Q satisfy (4.10). If $Q - P_m$ is nonconstant, write its degree as $j \geq 1$ and its leading coefficient as $a \neq 0$. For $j = 1$, its residue is exactly $-a$. For $j \geq 2$, expand the polynomial at $L_n + K(v) + \eta_n(v)$ in (4.11). The coefficient of L_n^j is constant in v, hence contributes no residue. The coefficient of L_n^{j-1} has residue $ja\operatorname{Res}_{v=0}(K + \eta_n) = -ja$. All remaining terms have degree at most $j-2$ in L_n and bounded residue coefficients. Hence

$$\operatorname*{Res}_{x=-n}(Q - P_m)(\psi(x)) = -ja(\log n)^{j-1} + O\big((1+\log n)^{j-2}\big). \tag{4.17}$$

Neither this expression nor the nonzero constant for $j = 1$ can tend to zero, contradicting the two normalization conditions. Thus $Q - P_m$ is constant. Conversely, an additive constant has zero residue and preserves the condition. □

The theorem establishes a uniqueness of functions, not an assertion about algebraic independence of Euler's constant or zeta values. It also explains why an unqualified uniqueness statement would be false: no residue condition can determine an additive constant.

Proposition 4.3 (Uniqueness of the complete Appell family)**.** *Suppose $A(t) \in \mathbb{C}[[t]]$, $A(0) = 1$, and $Q_m(Y) = m![t^m]A(t)e^{tY}$. If (4.10) holds for every $m \geq 1$, then $A(t) = \Gamma(1-t)$ as formal power series.*

Proof. By Theorem 4.2, $Q_m - P_m$ is constant for every m. If $A(t) - \Gamma(1-t)$ were nonzero, let $a_k t^k$, $a_k \neq 0$, be its first nonzero term; necessarily $k \geq 1$. The coefficient of t^{k+1} in $(A(t) - \Gamma(1-t))e^{tY}$ contains the nonconstant term $a_k Y$. This contradicts the conclusion for $m = k+1$. □

4.1 An explicit finite formula for the defect

Set $H_0^{(r)} = 0$, write $H_n = H_n^{(1)}$, and define

$$b_n = H_n - \gamma, \qquad a_{r,n} = H_n^{(r+1)} + (-1)^{r+1}\zeta(r+1) \quad (r \geq 1). \tag{4.18}$$

The exact convergent local expansion is

$$\psi(-n+v) = -\frac{1}{v} + b_n + \sum_{r\geq 1} a_{r,n}v^r, \qquad 0 < |v| < 1. \tag{4.19}$$

It follows by applying the recurrence to $\psi(v)$ and then using the Taylor expansion of $\psi(1+v)$.

Proposition 4.4 (Finite harmonic-polynomial defect formula)**.** *For every $m \geq 2$ and $n \geq 0$,*

$$r_{m,n} = m! \sum_{h=1}^{m} \frac{(-1)^h}{h!} [t^{m-h} v^{h-1}] \, \Gamma(1-t) \exp\left(t b_n + t \sum_{r=1}^{m-2} a_{r,n} v^r \right), \qquad \delta_{m,n} = r_{m,n} + m! e_{m-1}(n). \tag{4.20}$$

The sum over r is empty when $m = 2$. Formula (4.20) requires only finitely many coefficients of the displayed series; it is an exact polynomial computation, valid also at $n = 0$.

Proof. Substitute (4.19) in (4.3) and expand $e^{-t/v} = \sum_{h \geq 0} (-t)^h / (h! v^h)$. The residue from the term indexed by h is the coefficient of v^{h-1} in the holomorphic factor. Only $1 \leq h \leq m$ can contribute to $[t^m]$. Moreover, an occurrence of $a_{r,n} v^r$ consumes at least one additional power of t, so whenever such a coefficient occurs, $h \leq m - 1$ and $r \leq h - 1 \leq m - 2$. All higher $a_{r,n}$ can therefore be discarded before coefficient extraction. This proves the first equation; the second is the definition of the defect. □

For a transparent low-degree form, put

$$A_j(n) = \zeta(j) - H_n^{(j)}, \qquad j \geq 2. \tag{4.21}$$

The first defects are

$$\begin{aligned} \delta_{1,n} &= \delta_{2,n} = \delta_{3,n} = 0, \\ \delta_{4,n} &= -4A_3(n), \\ \delta_{5,n} &= -20 H_n A_3(n) + 5A_2(n)^2 - 20\zeta(2) A_2(n) + 25 A_4(n). \end{aligned} \tag{4.22}$$

These formulas can be checked without any infinite summation. For example, writing $h = H_n$ and $h_j = H_n^{(j)}$, the finite formula gives

$$\begin{aligned} r_{4,n} &= -4h^3 + 12hh_2 - 4h_3 - 4\zeta(3), \\ r_{5,n} &= -5h^4 + 30h^2 h_2 - 20hh_3 - 20\zeta(3)h - 10h_2^2 + 10\zeta(2)h_2 \\ &\quad + 5\zeta(2)^2 + 5h_4 - 25\zeta(4). \end{aligned} \tag{4.23}$$

Subtract the identities $-24e_3(n) = -4h^3 + 12hh_2 - 8h_3$ and $-120e_4(n) = -5h^4 + 30h^2 h_2 - 15h_2^2 - 40hh_3 + 30h_4$. The fifth defect then takes the form in (4.22) after $\zeta(2)^2 = 5\zeta(4)/2$ is used. In particular, scalar normalization is already exact at the level of residues through degree three. At degrees four and five the remaining terms are explicit decaying harmonic tails, rather than additional nondecaying lower-depth data.

4.2 Comparison with exact differential normalization

In the notation of the Gamma-polygamma construction, let

$$\mathcal{C}_m(x) = m! [t^m] \frac{\Gamma(1-t)\Gamma(x)}{\Gamma(x-t)}. \tag{4.24}$$

Taylor expansion of $\log \Gamma(x - t)$ gives the exact coefficient identity

$$\sum_{m \geq 0} \frac{\mathcal{C}_m(x)}{m!} t^m = \Gamma(1-t) e^{t\psi(x)} \exp\left(\sum_{r \geq 2} \frac{(-1)^{r+1} \psi^{(r-1)}(x)}{r!} t^r \right). \tag{4.25}$$

Thus the higher derivatives provide an exact correction to the scalar Appell family. The first examples are

$$\begin{aligned}
\mathcal{C}_2 &= P_2(\psi) - \psi', \\
\mathcal{C}_3 &= P_3(\psi) - 3P_1(\psi)\psi' + \psi'', \\
\mathcal{C}_4 &= P_4(\psi) - 6P_2(\psi)\psi' + 4P_1(\psi)\psi'' - \psi''' + 3(\psi')^2, \\
\mathcal{C}_5 &= P_5(\psi) - 10P_3(\psi)\psi' + 10P_2(\psi)\psi'' - 5P_1(\psi)\psi''' \\
&\quad + 15P_1(\psi)(\psi')^2 - 10\psi'\psi'' + \psi''''.
\end{aligned} \tag{4.26}$$

Indeed the quotient itself has only simple possible poles in x, and its exact residue is

$$\operatorname*{Res}_{x=-n} \frac{\Gamma(1-t)\Gamma(x)}{\Gamma(x-t)} = -t \prod_{j=1}^{n} \left(1 + \frac{t}{j}\right). \tag{4.27}$$

Coefficient extraction therefore yields

$$\operatorname*{Res}_{x=-n} \mathcal{C}_m(x) = -m! e_{m-1}(n), \qquad \operatorname*{Res}_{x=-n} \big(P_m(\psi(x)) - \mathcal{C}_m(x)\big) = \delta_{m,n}. \tag{4.28}$$

The scalar theorem cancels all nondecaying residue data. The differential family also cancels the explicit defects and every higher pole order. These are precise functional statements; they do not assert that the associated Laurent constants are arithmetically independent or form a minimal period basis.

4.3 A complete finite-part formula for inverse-Gamma derivatives

There is an all-degree form of the cancellation which explains the fifth-power identity before its raw-power endpoint corrections are computed. Define

$$\mathcal{A}_m(x) = (-1)^m \Gamma(x) \frac{d^m}{dx^m} \frac{1}{\Gamma(x)}, \qquad \frac{\Gamma(x)}{\Gamma(x-t)} = \sum_{m \geq 0} \mathcal{A}_m(x) \frac{t^m}{m!}. \tag{4.29}$$

Then $\mathcal{A}_1 = \psi$, $\mathcal{A}_2 = \psi^2 - \psi'$, and $\mathcal{A}_{m+1} = \psi \mathcal{A}_m - \mathcal{A}_m'$.

Corollary 4.5 (All-degree differential finite parts)**.** *For $m \geq 1$,*

$$\boxed{\operatorname{FP} \int_0^1 \mathcal{A}_m(x)\, dx = m![t^m] \frac{\sin(\pi t)}{\pi t} - m! \sum_{d=1}^{m-1} (-1)^{m-d-1} c_{m-d-1} \lambda_d.} \tag{4.30}$$

The sum is empty for $m = 1$. For odd m the first term is zero; for $m = 2k$ it is $(-1)^k \pi^{2k}/(2k+1)$.

Proof. Multiplying the series for $\mathcal{C}_m$ by $1/\Gamma(1-t)$ gives

$$\mathcal{A}_m = \sum_{k=0}^{m} \frac{m!}{k!} (-1)^{m-k} c_{m-k} \mathcal{C}_k.$$

Finite parts are linear. The degree-zero integral is one, and $\operatorname{FP} \int_0^1 \mathcal{C}_1 = \gamma$ because $\int_\varepsilon^1 \psi = -\log \Gamma(\varepsilon)$ has constant term zero. For $k \geq 2$ use Theorem 3.4. The terms containing only c_k combine to

$$m![t^m] \frac{1}{\Gamma(1-t)\Gamma(1+t)} = m![t^m] \frac{\sin(\pi t)}{\pi t},$$

and the remaining terms are exactly the displayed finite sum. □

In degree five this reads

$$\begin{aligned}
\operatorname{FP} \int_0^1 \mathcal{A}_5(x)\, dx = &-120\xi + 120\gamma\theta - 60(\gamma^2 - \zeta(2))\eta \\
&+ 20(\gamma^3 - 3\gamma\zeta(2) + 2\zeta(3))\lambda_1.
\end{aligned} \tag{4.31}$$

The next section proves exactly how $\mathcal{A}_5$ differs, after integration with the specified finite parts, from the scalar power ψ^5. The resulting additional mixed integral is an analytic quantity, not an omitted integration constant.

5 The fifth digamma power and its mixed coordinate

This section resolves the fifth-power calculation posed in the inspected quartic report. It fixes all endpoint constants and identifies the additional mixed spectral combination which occurs beside the height-one Laurent coefficients. The result does not assert that this mixed combination is arithmetically independent of the height-one data.

Write $z_j = \zeta(j)$ and $\gamma = \gamma_0$. Strict multiple zeta functions have decreasing indices:

$$Z_r(s_1, \dots, s_r) = \sum_{n_1 > \cdots > n_r \geq 1} n_1^{-s_1} \cdots n_r^{-s_r}.$$

For their height-one germs put

$$\lambda_r = [u^1] Z_r(1 + u, \underbrace{1, \dots, 1}_{r-1}), \qquad \lambda_1 = -\gamma_1, \quad \lambda_2 = \eta, \quad \lambda_3 = \theta, \quad \lambda_4 = \xi. \tag{5.1}$$

The bracket extracts the coefficient itself, with no factorial or sign. The coordinate finite part of an integral is the constant coefficient of its cutoff expansion in ε for the integral from ε to the stated upper endpoint.

5.1 The identities

Define

$$R(s) = Z_2(s, 3) - z_3 \zeta(s), \tag{5.2}$$

$$J(s) = 2R(s) + 2Z_2(s + 2, 1) + Z_2(s + 1, 2) - 2\gamma\zeta(s + 2) + z_2\zeta(s + 1). \tag{5.3}$$

Here R is holomorphic in $\operatorname{Re} s > -1$, and J is holomorphic in $\operatorname{Re} s > 0$. In particular $J'(1)$ is an ordinary derivative of a holomorphic function, not a coefficient requiring a new choice of finite-part normalization.

Theorem 5.1 (Fifth-power and mixed-polygamma bridges)**.** *Set*

$$Q_5 = \operatorname{FP} \int_0^1 \psi(x)^5 \, dx, \qquad I = \operatorname{FP} \int_0^1 \psi(x)\psi'(x)^2 \, dx.$$

Then

$$\boxed{I = -2\gamma z_3 + \frac{35}{6} z_4 + 2J'(1).} \tag{5.4}$$

The fifth-power identity is

$$\boxed{\begin{aligned} Q_5 = {} & 30\gamma^2 z_2 - 60\gamma z_3 + \frac{35}{3} z_4 - 120\xi + 120\gamma\theta \\ & - 60(\gamma^2 - z_2)\eta + 20(\gamma^3 - 3\gamma z_2 + 2z_3)\lambda_1 + 10J'(1). \end{aligned}} \tag{5.5}$$

Equivalently, the mixed integral itself isolates the extra coordinate:

$$\boxed{\begin{aligned} Q_5 - 5I = {} & 30\gamma^2 z_2 - 50\gamma z_3 - \frac{35}{2} z_4 - 120\xi + 120\gamma\theta \\ & - 60(\gamma^2 - z_2)\eta + 20(\gamma^3 - 3\gamma z_2 + 2z_3)\lambda_1. \end{aligned}} \tag{5.6}$$

All the ordinary zeta derivatives in the compact notation are explicit:

$$J'(1) = 2R'(1) + 2 \left. \frac{\partial}{\partial s} Z_2(s, 1) \right|_{s=3} + \left. \frac{\partial}{\partial s} Z_2(s, 2) \right|_{s=2} - 2\gamma\zeta'(3) + z_2\zeta'(2). \tag{5.7}$$

If $\phi_{13} = [u^1] Z_2(1 + u, 3)$, then $R'(1) = \phi_{13} + z_3\gamma_1 = \phi_{13} - z_3\lambda_1$.

The last conversion matters. Substituting ϕ_{13} for $R'(1)$ without its $z_3\gamma_1$ correction would give an incorrect fifth-power formula. The mixed term is one specified linear combination. No claim that its separate components form a minimal arithmetic basis is needed.

5.2 A convergent arithmetic form and the spectral germs

For $n \geq 1$ use

$$b = \psi(n), \qquad c = 2z_2 - \psi'(n), \qquad d = \frac{1}{2}\psi''(n), \qquad L_n = \log n, \tag{5.8}$$

where b, c, d in a summand always refer to that summand's n. Then

$$J(s) = \sum_{n\geq 1} \frac{1}{n^s}\left(2d + \frac{c}{n} + \frac{2b}{n^2}\right), \qquad \operatorname{Re} s > 0, \tag{5.9}$$

$$I = -2\gamma z_3 + \frac{35}{6}z_4 - \sum_{n\geq 1}\left(\frac{4d}{n} + \frac{2c}{n^2} + \frac{4b}{n^3}\right)L_n. \tag{5.10}$$

Another expression for Q_5 is

$$\begin{aligned} Q_5 = {} & 30\gamma^2 z_2 - 60\gamma z_3 + \frac{35}{3}z_4 + 20\gamma_4 - 120 z_2\gamma_2 \\ & + 20\sum_{n\geq 1}\frac{(b^3 - 3bc + d - L_n^3 + 6z_2 L_n)L_n}{n} \\ & - 20\sum_{n\geq 1}\frac{bL_n}{n^3} - 10\sum_{n\geq 1}\frac{cL_n}{n^2}. \end{aligned} \tag{5.11}$$

Every series in these displays converges absolutely. Indeed $\psi(n) = \log n + O(n^{-1})$, $\psi'(n) = O(n^{-1})$, and $\psi''(n) = O(n^{-2})$. The summand in the first sum of (5.11) is $O((1+\log^3 n)/n^2)$. These estimates also prove normal convergence of (5.9) and of each fixed spectral derivative on compact subsets of its half-plane. The series for R is

$$R(s) = \sum_{n\geq 1}\frac{H_{n-1}^{(3)} - z_3}{n^s};$$

its summands are $O(n^{-\operatorname{Re} s - 2})$.

The height-one Laurent germs used here have a direct existence proof. For $r \geq 1$ and $\operatorname{Re} s > 1$,

$$Z_r(s, 1^{r-1}) = \frac{(-1)^{r-1}}{(r-1)!\,\Gamma(s)}\int_0^\infty \frac{t^{s-1}\log^{r-1}(1-e^{-t})}{e^t - 1}\,dt. \tag{5.12}$$

This follows from the elementary-symmetric generating series and termwise Mellin integration. Near zero the kernel is its displayed leading term $(-1)^{r-1}\log^{r-1} t/((r-1)!t)$ plus $O(1+|\log t|^{r-1})$. Subtraction of that term gives

$$\Gamma(1+u)Z_r(1+u, 1^{r-1}) = u^{-r} + V_r(u), \qquad V_r(0) = 0,$$

with V_r holomorphic near zero. The assertion $V_r(0) = 0$ follows by integrating the derivative of $\log^r(1-e^{-t})$ and comparing its lower boundary with $\log^r t$. Thus the principal and constant coefficients are fixed by $1/\Gamma(1+u)$, and in particular ξ in (5.1) is well defined.

5.3 Global partial fractions with a fixed entire remainder

For $n \geq 0$, set

$$\begin{aligned} b_n &= H_n - \gamma, & c_n &= z_2 + H_n^{(2)}, \\ d_n &= H_n^{(3)} - z_3, & e_n &= z_4 + H_n^{(4)}. \end{aligned} \tag{5.13}$$

The local digamma expansion is

$$\psi(-n+\varepsilon) = -\varepsilon^{-1} + b_n + c_n\varepsilon + d_n\varepsilon^2 + e_n\varepsilon^3 + O(\varepsilon^4).$$

For $F_5(z) = \psi(z)^5$ and $F_I(z) = \psi(z)\psi'(z)^2$, let

$$P_{X,n}(z) = \sum_{j=1}^{5} \frac{p_{X,j,n}}{(z+n)^j}, \qquad X \in \{5, I\}.$$

The principal coefficients, with subscripts n suppressed in the table, are

j	$p_{5,j}$	$p_{I,j}$
5	-1	-1
4	$5b$	b
3	$-10b^2 + 5c$	$-c$
2	$10b^3 - 20bc + 5d$	$2bc - 3d$
1	$-5b^4 + 30b^2c - 20bd - 10c^2 + 5e$	$4bd + c^2 - 5e$

Write $r_{X,n} = p_{X,1,n}$. The constant coefficients at zero are

$$\kappa_5 = -\gamma^5 + 20\gamma^3 z_2 - 30\gamma^2 z_3 - 30\gamma z_2^2 + 20\gamma z_4 + 20 z_2 z_3 - 5 z_5, \tag{5.14}$$

$$\kappa_I = -\gamma z_2^2 - 6\gamma z_4 - 2 z_2 z_3 + 7 z_5. \tag{5.15}$$

Lemma 5.2 (Two normalized Mittag–Leffler expansions)**.** *For $X = 5, I$ and $z \notin \{0, -1, -2, \ldots\}$,*

$$F_X(z) = P_{X,0}(z) + \kappa_X + \sum_{n \geq 1} (P_{X,n}(z) - P_{X,n}(0)). \tag{5.16}$$

The series converges normally on compact sets avoiding the poles.

Proof. The coefficient table follows by finite multiplication of the local digamma series and its derivative. For the entire remainder, put $f_X = F_X - P_{X,0}$ and integrate $z f_X(w)/(w(w-z))$ on $|w| = N + 1/2$. Digamma asymptotics in the right half-plane and the reflection formula in the left half-plane give $f_X(w) = O(1 + \log^5 N)$ on these circles. The cotangent and its fixed derivatives are uniformly bounded there: away from the real axis use their exponential expressions, and on the remaining arcs the real coordinate stays a fixed distance from the integers. The contour integral is therefore $O((1 + \log^5 N)/N)$ and tends to zero. Its residues are $f_X(z)$, $-f_X(0)$, and $-(P_{X,n}(z) - P_{X,n}(0))$. This proves the identity. Finally $p_{5,j,n} = O(1 + \log^{5-j} n)$ and the corresponding mixed coefficients have no greater growth. The two-point subtractions give an additional inverse power of n, proving normal convergence. □

5.4 The finite summation ledger

Integrating (5.16) and telescoping each pole of order at least two gives

$$\mathrm{FP} \int_0^1 F_X(x)\, dx = B_X + \sum_{n \geq 1} r_{X,n} \left(\log\left(1 + \frac{1}{n}\right) - \frac{1}{n} \right), \tag{5.17}$$

where

$$B_X = \kappa_X + \sum_{j=2}^{5} \sum_{n \geq 1} \left(\frac{p_{X,j,n} - p_{X,j,n-1}}{(j-1)n^{j-1}} - \frac{p_{X,j,n}}{n^j} \right). \tag{5.18}$$

This formula includes the finite parts at the origin. For example, the lower endpoint contribution of $p_{X,j,0} x^{-j}$ is $-p_{X,j,0}/(j-1)$; it cancels the initial term of that pole's finite telescoping sum. The simple-pole finite part is zero.

Here is a shorter directly reproducible form of all the nonlogarithmic terms. In the sums in this display only, let $b = b_{n-1}$, $c = c_{n-1}$, $d = d_{n-1}$. Then

$$B_5 = \kappa_5 + \sum_{n \geq 1} \left\{ \frac{-10b^3 + 20bc - 5d + 30b^2 - 20c}{n^2} + \frac{-20b^2 + 15c}{n^3} + \frac{5b - 35/6}{n^4} + \frac{6}{n^5} \right\}, \tag{5.19}$$

$$B_I = \kappa_I + \sum_{n \geq 1} \left\{ \frac{-2bc + 3d + 2c}{n^2} + \frac{2b - c}{n^3} + \frac{-3b - 7/6}{n^4} + \frac{2}{n^5} \right\}. \tag{5.20}$$

All these sums converge absolutely.

To evaluate them and the finite residue sums, write

$$H_N(a_1,\dots,a_d)=\sum_{N\ge n_1>\cdots>n_d\ge 1} n_1^{-a_1}\cdots n_d^{-a_d},\qquad e_j(N)=H_N(1^j).$$

The finite product rule is the quasi-shuffle rule with a positive merged index. The summation rule is equally elementary:

$$\sum_{n=1}^{N}\frac{H_n(a_1,\dots,a_d)}{n^p}=H_N(p,a_1,\dots,a_d)+H_N(p+a_1,a_2,\dots,a_d),\tag{5.21}$$

with the second term absent for an empty word. Both statements follow by separating strict inequalities from equalities of the outer indices. There is no infinite rearrangement in either identity.

For completeness, the convergent values required in this section are

value	reduction
$Z_2(2,1)$	z_3
$Z_2(3,1),\ Z_2(2,2),\ Z_3(2,1,1)$	$z_4/4,\ 3z_4/4,\ z_4$
$Z_2(4,1),\ Z_3(3,1,1)$	$2z_5-z_2z_3$
$Z_2(3,2),\ Z_3(2,2,1)$	$3z_2z_3-11z_5/2$
$Z_2(2,3),\ Z_3(2,1,2)$	$9z_5/2-2z_2z_3$
$Z_4(2,1,1,1)$	z_5

(5.22)

These are the classical low-weight reductions. Their use here can be checked from the finite product rule, the depth-two sum theorem and iterated-integral duality. At weight five, for example, the stuffle and shuffle products are respectively

$$z_2z_3=Z_2(2,3)+Z_2(3,2)+z_5,\qquad z_2z_3=Z_2(2,3)+3Z_2(3,2)+6Z_2(4,1).$$

Together with $Z_2(4,1)+Z_2(3,2)+Z_2(2,3)=z_5$, these give the three depth-two weight-five entries. Substitution $t\mapsto 1-t$ in the iterated integrals gives the paired dual values in the table. The same arguments at weights three and four give the first two rows; $z_2^2=5z_4/2$ is used throughout.

Here is an explicit finite identity which records the full fifth-power constant. In this display set $h_j=H_N^{(j)}$, $h=h_1$, $E_j=e_j(N)$, and abbreviate $H_N(a_1,\dots,a_d)$ by $H(a_1,\dots,a_d)$. Put

$$A=\gamma^3-3\gamma z_2+z_3,\qquad r_0=-5\gamma^4+30\gamma^2z_2-20\gamma z_3-10z_2^2+5z_4.$$

Then the exact equality for every $N\ge 1$ is

$$\begin{aligned}\sum_{n=1}^{N}\frac{r_{5,n}}{n}=\ &-120E_5+120\gamma E_4+60(z_2-\gamma^2)E_3+20AE_2+r_0h+20Ah_2\\&-120H(2,1,1,1)+120\gamma H(2,1,1)+(50z_2-60\gamma^2)H(2,1)\\&+20E_2h_3-20H(3,1,1)-20hh_4+20H(4,1)+20h_5\\&+10hH(2,2)-10H(2,1,2)-10H(2,2,1)+10H(2,3)\\&-20\gamma hh_3+20\gamma H(3,1)+10z_2hh_2.\end{aligned}\tag{5.23}$$

One derivation first writes the residue polynomial as

$$\begin{aligned}r_{5,n}=\ &-120e_4(n)+120\gamma e_3(n)+60(z_2-\gamma^2)e_2(n)+20AH_n+r_0\\&+20(H_n-\gamma)H_n^{(3)}+5\big(H_n^{(2)}\big)^2-25H_n^{(4)}+10z_2H_n^{(2)}.\end{aligned}$$

Apply (5.21) to every term, and then use the finite product rule to collect the products occurring in (5.23).

For the mixed residue there is a shorter identity. Put $C_0 = 4\gamma z_3 + z_2^2 - 5z_4$. Then

$$\begin{aligned}\sum_{n=1}^{N} \frac{r_{I,n}}{n} = 4E_2(h_3 - z_3) + h\{2H(2,2) - 4h_4 - 4\gamma h_3 + 2z_2h_2 + C_0\} \\ - 4H(3,1,1) - 2H(2,1,2) - 2H(2,2,1) + 2H(2,3) \\ + 4H(4,1) + 4h_5 + 4\gamma H(3,1) - 4z_3h_2 - 2z_2H(2,1). \end{aligned} \tag{5.24}$$

This is again a finite product-and-prefix calculation.

Lemma 5.3 (All finite constants)**.** *As $N \to \infty$, with $L = \log N$,*

$$\sum_{n=1}^{N} \frac{r_{5,n}}{n} = -L^5 + 20z_2L^3 - 90z_4L + C_5 + o(1), \tag{5.25}$$

$$r_{5,N} \log(N+1) = -5L^5 + 60z_2L^3 - 90z_4L + o(1), \tag{5.26}$$

where

$$C_5 = -\gamma^5 + 30\gamma^3 z_2 - 80\gamma^2 z_3 - \frac{65}{2}\gamma z_4 + 60 z_2 z_3 - 69 z_5. \tag{5.27}$$

For the mixed residue,

$$\sum_{n\geq 1} \frac{r_{I,n}}{n} = C_I = \gamma z_4 - 12 z_2 z_3 + 15 z_5, \qquad r_{I,N} \log(N+1) = o(1). \tag{5.28}$$

Finally the nonlogarithmic terms satisfy

$$B_5 - C_5 = 30\gamma^2 z_2 - 60\gamma z_3 + \frac{35}{3} z_4, \qquad B_I - C_I = -2\gamma z_3 + \frac{35}{6} z_4. \tag{5.29}$$

Proof. In (5.23), substitute $h = \log N + \gamma + O(N^{-1})$ and $h_j = z_j + O(N^{1-j})$ for $j \geq 2$, and replace the words with outer exponent at least two by their limits in (5.22). Every discarded term is $o(1)$ even after multiplication by the displayed powers of $\log N$. The elementary polynomials are fixed, without any interpolation, by

$$E_0 = 1, \qquad jE_j = \sum_{r=1}^{j} (-1)^{r-1} h_r E_{j-r}.$$

This gives (5.25) including its constant (5.27). For (5.26), directly substitute $b_N = L + O(N^{-1})$, $c_N = 2z_2 + O(N^{-1})$, $d_N = O(N^{-2})$, and $e_N = 2z_4 + O(N^{-3})$ in the residue. Its constant polynomial term is $-40z_2^2 + 10z_4 = -90z_4$.

For (5.24), the first term tends to zero. The expression in braces multiplying h has zero limit by $Z_2(2,2) = 3z_4/4$ and $z_2^2 = 5z_4/2$; its convergence is fast enough that its product with h tends to zero. The remaining convergent words have value

$$\begin{aligned} - 4Z_3(3,1,1) - 2Z_3(2,1,2) - 2Z_3(2,2,1) + 2Z_2(2,3) \\ + 4Z_2(4,1) + 4z_5 + 4\gamma Z_2(3,1) - 6z_2z_3 \\ = \gamma z_4 - 12 z_2 z_3 + 15 z_5. \end{aligned}$$

Also $r_{I,n} = O(n^{-1}) + O(\log n/n^2)$, since its prospective constant term $4z_2^2 - 10z_4$ vanishes. This proves the ordinary convergence and the boundary assertion in (5.28). Finally, expand the products in (5.19) and (5.20) by the finite quasi-shuffle rule, prefix their outer denominator, and use (5.22). The results are respectively C_5 and C_I plus the two explicitly displayed corrections in (5.29). □

The preceding calculation is a full constant-term computation. In particular it retains the $-90z_4 \log N$ terms on both sides of the summation-by-parts boundary; neither term may be dropped before their cancellation.

5.5 Identification of the logarithmic terms

The residue differences are particularly short. For $n \geq 1$, with $b = b_{n-1}$, $c = c_{n-1}$, $d = d_{n-1}$,

$$r_{5,n} - r_{5,n-1} = -\frac{20(b^3 - 3bc + d)}{n} + \frac{20b}{n^3} + \frac{10c}{n^2}, \tag{5.30}$$

$$r_{I,n} - r_{I,n-1} = \frac{4d}{n} + \frac{4b}{n^3} + \frac{2c}{n^2}. \tag{5.31}$$

For any residue sequence, finite summation by parts gives

$$\sum_{n=1}^{N} r_{X,n} \log\left(1 + \frac{1}{n}\right) = r_{X,N} \log(N+1) - \sum_{n=1}^{N} (r_{X,n} - r_{X,n-1}) \log n. \tag{5.32}$$

For I, equations (5.17), (5.28), (5.29), and (5.31) immediately give (5.10) and hence (5.4).

For Q_5, define

$$M_N = \sum_{n=1}^{N} \frac{(b_{n-1}^3 - 3b_{n-1}c_{n-1} + d_{n-1}) \log n}{n}.$$

Equations (5.25)–(5.32) give

$$\begin{aligned} Q_5 = B_5 - C_5 + 20 \lim_{N\to\infty} \left(M_N - \frac{\log^5 N}{5} + 2z_2 \log^3 N \right) \\ - 20 \sum_{n\geq 1} \frac{b_{n-1} \log n}{n^3} - 10 \sum_{n\geq 1} \frac{c_{n-1} \log n}{n^2}. \end{aligned} \tag{5.33}$$

Splitting off $\log^3 n - 6z_2 \log n$ and using the ordinary Stieltjes limits gives (5.11).

For the exact spectral form introduce, initially for $\operatorname{Re} s > 1$,

$$T(s) = \sum_{n\geq 1} \frac{b_{n-1}^3 - 3b_{n-1}c_{n-1} + d_{n-1}}{n^s}.$$

Finite elementary-symmetric algebra gives

$$\begin{aligned} T(s) = 6Z_4(s,1,1,1) - 6\gamma Z_3(s,1,1) + 3(\gamma^2 - z_2)Z_2(s,1) \\ - (\gamma^3 - 3\gamma z_2 + 2z_3)\zeta(s) - R(s). \end{aligned} \tag{5.34}$$

Moreover $T(s) + \zeta'''(s) - 6z_2\zeta'(s)$ is holomorphic in $\operatorname{Re} s > 0$. Thus T has principal part $6u^{-4} - 6z_2u^{-2}$ at $s = 1 + u$, and the finite limit in (5.33) is exactly $-[u^1]T(1+u)$. One can see the sign directly by differentiating the normally convergent subtracted Dirichlet series; the ordinary limits contribute $\gamma_4 - 6z_2\gamma_2$. By (5.34),

$$[u^1]T(1+u) = 6\xi - 6\gamma\theta + 3(\gamma^2 - z_2)\eta - (\gamma^3 - 3\gamma z_2 + 2z_3)\lambda_1 - R'(1).$$

The two remaining sums in (5.33) are

$$-\frac{\partial}{\partial s} Z_2(s,1)\Big|_{s=3} + \gamma\zeta'(3), \qquad -\frac{\partial}{\partial s} Z_2(s,2)\Big|_{s=2} - z_2\zeta'(2),$$

respectively. Substitution proves (5.5); subtraction of five times (5.4) gives (5.6).

5.6 A differential-polynomial interpretation and exact endpoints

The fifth inverse-Gamma differential polynomial is

$$\mathcal{A}_5 = \psi^5 - 10\psi^3\psi' + 15\psi(\psi')^2 + 10\psi^2\psi'' - 10\psi'\psi'' - 5\psi\psi''' + \psi''''. \tag{5.35}$$

It equals $-\Gamma\, D^5(1/\Gamma)$. Let $\Delta G = G(1) - [x^0]G(x)$ for a Laurent-meromorphic function at zero. Coordinate finite parts obey $\mathrm{FP}\int_0^1 G'(x)\,dx = \Delta G$. The complete endpoint table needed for (5.35) is

G	ΔG
ψ^4	$12\gamma^2 z_2 - 12\gamma z_3 - 11 z_4$
$\psi^2\psi'$	$2\gamma z_3 + 3z_4/2$
$(\psi')^2$	$-6z_4$
$\psi\psi''$	$8z_4$
ψ'''	0

(5.36)

Integration by parts, with these constants retained, gives

$$Q_5 - 5I = \mathrm{FP}\int_0^1 \mathcal{A}_5(x)\,dx + 30\gamma^2 z_2 - 50\gamma z_3 - \frac{35}{2} z_4. \tag{5.37}$$

Consequently (5.6) is compatible with the all-degree inverse-Gamma construction, while the raw fifth power contains I. This distinguishes a proved differential-polynomial cancellation from the separate proposed cancellation by a scalar polynomial in ψ. The formulas here neither prove nor disprove the existence of a further ordinary-constant identity eliminating $J'(1)$.

5.7 Normalized primitives and the polygamma hierarchy

For $n \geq 1$, define on $\mathrm{Re}\, z > 0$

$$F_1(n,z) = \log\left(1 + \frac{z}{n}\right) - \frac{z}{n}, \qquad F_j(n,z) = \frac{(n+z)^{1-j} - n^{1-j}}{1-j} - \frac{z}{n^j} \quad (j \geq 2),$$

with the logarithm real for positive arguments.

Corollary 5.4 (Two normalized primitives)**.** *For $X \in \{5, I\}$ the expression*

$$\mathcal{P}_X(z) = r_{X,0}\log z + \sum_{j=2}^{5} \frac{p_{X,j,0} z^{1-j}}{1-j} + \kappa_X z + \sum_{n\geq 1}\sum_{j=1}^{5} p_{X,j,n} F_j(n,z) \tag{5.38}$$

satisfies $\mathcal{P}_X'(z) = F_X(z)$ and has zero constant term at the singular origin. For real $b > 0$,

$$\mathrm{FP}\int_0^b F_X(x)\,dx = \mathcal{P}_X(b).$$

For $0 < a < b$ the ordinary integral is $\mathcal{P}_X(b) - \mathcal{P}_X(a)$.

Proof. Each F_j is the primitive vanishing at zero of $(n+z)^{-j} - n^{-j}$. On compact sets it is $O(n^{-j-1})$; the coefficient bounds from Lemma 5.2 therefore justify normal convergence and differentiation. The series converges uniformly on each disk $|z| \leq r < 1$, where all its summands vanish at zero. The explicitly displayed singular terms and logarithm consequently leave no constant term. □

Corollary 5.5 (All fixed polygamma orders)**.** *For every integer $r \geq 1$, away from the poles,*

$$D^r F_X(z) = (-1)^r r! \sum_{n\geq 0}\sum_{j=1}^{5} \binom{r+j-1}{j-1} \frac{p_{X,j,n}}{(n+z)^{r+j}}. \tag{5.39}$$

The series converges normally. The left sides are explicitly

$$D^r F_5 = \sum_{r_1+\cdots+r_5=r} \frac{r!}{r_1!\cdots r_5!} \prod_{i=1}^{5} \psi^{(r_i)}, \qquad D^r F_I = \sum_{r_1+r_2+r_3=r} \frac{r!}{r_1! r_2! r_3!} \psi^{(r_1)} \psi^{(r_2+1)} \psi^{(r_3+1)}.$$

At $z = 1$ these are finite polynomials in γ and ordinary zeta values, using $\psi^{(j)}(1) = (-1)^{j+1} j! \zeta(j+1)$ for $j \geq 1$.

Proof. Differentiate (5.16). For $r \geq 1$ all constant subtractions disappear, and the summands are bounded on compact sets by a constant times $n^{-r-1}(1 + \log^4 n)$. The product rule gives the displayed left sides. □

5.8 Verification and scope

The accompanying exact replay checks 87 identities: principal coefficients, constant Laurent terms, residue differences, both finite harmonic summation formulas, all finite asymptotic constants, nonlogarithmic reductions, spectral extraction, endpoint terms, primitive derivatives, and the derivative kernels through order eight. The finite harmonic checks are symbolic identities for arbitrary N; they are not a fit to a finite sample of integers. As regression checks, the same summation code recovers the previously proved cubic and quartic constants.

Independent numerical evaluations give

$$Q_5 = -8.690303962624538266295589662365653576770293544923884292 66\ldots,$$
$$I = -1.019465821941729267443698213915546137467423003511484751 73\ldots.$$

One route integrates the convergent Taylor series of the functions after their fifth-order singularity has been factored out near zero, and performs ordinary quadrature away from zero. Cutoffs $1/8$ and $1/16$ agree to about 77 decimal digits. A second route uses (5.10) and (5.11), summing a finite prefix and expressing a truncated digamma asymptotic tail through Hurwitz-zeta derivatives. At the finest recorded settings the differences from the direct calculation are below $1.02 \cdot 10^{-48}$ and $3.83 \cdot 10^{-48}$, respectively. These are floating-point cross-checks, not interval-certified enclosures. The exact identities rest on the proofs above.

The advance relative to the inspected quartic report is the complete fifth-power bridge, its mixed-coordinate identification, all finite corrections, and the normalized primitive and derivative consequences. The classical partial-fraction and multiple-zeta tools are not claimed as new. The scalar-polynomial elimination question remains distinct: the presently proved cancellation naturally uses the additional integral I or, equivalently, the differential polynomial $\mathcal{A}_5$.

6 Polylogarithmic kernels, derivatives, and normalized primitives

The residue normalization has useful consequences beyond one finite part. It gives convergent antiderivatives, strict multiple-zeta Taylor coefficients, and explicit polylogarithmic kernels. These are exact identities of functions on stated domains. The Gamma and polylogarithm ingredients are classical; the point here is that the same normalization works for the ordinary integral, the poles, and all subsequent differentiations.

6.1 A primitive with a fixed additive constant

Theorem 6.1 (The normalized Gamma-polygamma primitive)**.** *Let $m \geq 2$ and $\Omega_m = \mathbb{C} \setminus (-\infty, -(m-1)]$. With the principal logarithm, put*

$$\mathfrak{P}_m(z) = -m!\zeta(m)z - m! \sum_{n \geq m-1} e_{m-1}(n) \left\{ \operatorname{Log}\left(1 + \frac{z}{n}\right) - \frac{z}{n} \right\}. \tag{6.1}$$

Then $\mathfrak{P}_m$ is holomorphic on Ω_m, $\mathfrak{P}_m(0) = 0$, and $\mathfrak{P}'_m = \mathcal{C}_m$. In particular

$$\mathfrak{P}_m(1) = m!(c_m - \lambda_{m-1}). \tag{6.2}$$

For any real $-(m-1) < a < b$, the ordinary integral of $\mathcal{C}_m$ is $\mathfrak{P}_m(b) - \mathfrak{P}_m(a)$.

Proof. On every fixed compact set the bracket in (6.1) is $O(n^{-2})$, uniformly in z. The logarithm is holomorphic on the stated slit because the first nonzero residue occurs at $n = m-1$. The normal convergence and the same bound for the differentiated bracket justify termwise differentiation, giving (3.12). Every term vanishes at zero. Theorem 3.4 gives the value at one. □

6.2 The all-ones polylogarithm as a kernel

The harmonic generating identity is

$$\sum_{n\geq 1} e_d(n)y^n = \frac{\mathrm{Li}_{\{1\}^d}(y)}{1-y} = \frac{(-\log(1-y))^d}{d!(1-y)}, \qquad |y| < 1. \tag{6.3}$$

The factor $(1-y)^{-1}$ records the inclusive upper cutoff n. Omitting it changes the harmonic convention and the resulting poles.

Corollary 6.2 (Polylogarithmic integral and primitive kernels)**.** *For $m \geq 2$ and $\operatorname{Re} z > -(m-1)$,*

$$\mathcal{C}_m(z) = -m!\zeta(m) + m!\int_0^1 (1-y^z)\frac{\mathrm{Li}_{\{1\}^{m-1}}(y)}{y(1-y)}\,dy, \tag{6.4}$$

$$\mathfrak{P}_m(z) = -m!\zeta(m)z + m!\int_0^1 \left\{z - \frac{y^z-1}{\log y}\right\}\frac{\mathrm{Li}_{\{1\}^{m-1}}(y)}{y(1-y)}\,dy. \tag{6.5}$$

Here $y^z = e^{z\log y}$ uses the real logarithm for $0 < y < 1$. Both integrals are ordinary absolutely convergent integrals.

Proof. Insert $(n+z)^{-1} - n^{-1} = \int_0^1 y^{n-1}(y^z-1)\,dy$ into (3.12) and use (6.3). Near zero the polylogarithm is $O(y^{m-1})$, giving precisely the half-plane in the statement. Near one, $1-y^z = O(1-y)$; the remaining logarithmic power is integrable. These estimates are uniform on compact parameter sets and justify summation and differentiation. Integrating with respect to z from zero gives the second formula and its normalization. □

6.3 All derivatives and the exact Taylor radius

Corollary 6.3 (Derivative Euler sums)**.** *For every $m \geq 2$ and integer $r \geq 1$,*

$$\mathcal{C}_m^{(r)}(z) = (-1)^{r+1}m!r!\sum_{n\geq m-1}\frac{e_{m-1}(n)}{(n+z)^{r+1}}. \tag{6.6}$$

The series converges normally away from the actual poles. At zero,

$$\mathcal{C}_m^{(r)}(0) = (-1)^{r+1}m!r!\left\{Z_m(r+1,\{1\}^{m-1}) + Z_{m-1}(r+2,\{1\}^{m-2})\right\}. \tag{6.7}$$

Equivalently, the Taylor series is

$$\mathcal{C}_m(z) = -m!\zeta(m) + m!\sum_{r\geq 1}(-1)^{r+1}\left\{Z_m(r+1,\{1\}^{m-1}) + Z_{m-1}(r+2,\{1\}^{m-2})\right\}z^r. \tag{6.8}$$

Its radius of convergence is exactly $m-1$.

Proof. Differentiate the normally convergent partial fractions. The bound $e_{m-1}(n) = O((1 + \log n)^{m-1})$ is sufficient for every $r \geq 1$. The exact finite identity $e_d(n) = e_d(n-1) + e_{d-1}(n-1)/n$ separates the inclusive cutoff into two strict sums, proving (6.7). The Taylor coefficients follow by division by $r!$. Finally the origin is removable and the nearest actual singularity is the simple pole at $-(m-1)$, whose residue is nonzero by Theorem 3.2; this proves the exact radius. □

For example $\mathcal{C}_2'(0) = 4\zeta(3)$ and $\mathcal{C}_3'(0) = 15\zeta(4)/2$. These recover the same coefficients by differentiating the Bell-polygamma expressions. More generally, every derivative at zero becomes a convergent multiple-zeta value, whereas integration over one unit interval produces a regular Laurent coefficient. This provides a direct functional connection between the two types of constants.

On $\operatorname{Re} z > -(m-1)$ the derivative also has the kernel

$$\mathcal{C}_m^{(r)}(z) = (-1)^{r+1} m! \int_0^1 y^z (-\log y)^r \frac{\operatorname{Li}_{\{1\}^{m-1}}(y)}{y(1-y)}\, dy. \tag{6.9}$$

The powers of $\log y$ therefore represent spatial differentiation, while the logarithmic Mellin powers in (2.5) represent spectral differentiation.

The Gamma recurrence also supplies the exact discrete relation

$$\mathcal{C}_m(x+1) - \mathcal{C}_m(x) = \frac{m}{x}\mathcal{C}_{m-1}(x+1), \qquad m \geq 1. \tag{6.10}$$

It follows by taking coefficients in $G(x+1,t) = xG(x,t)/(x-t)$. The identity holds meromorphically; removable values at exceptional points are obtained by continuation. At positive integers it is the classical complete-harmonic recurrence.

6.4 Antiderivatives in the spectral parameter

The normalized germ Λ_d from Corollary 2.2 has a convergent primitive as well. On $\operatorname{Re} u > -1$, define

$$\mathfrak{S}_d(u) = \frac{1}{d!} \int_0^\infty \frac{V_d(v)}{v} \frac{(u \log v - 1)v^u + 1}{(\log v)^2}\, dv. \tag{6.11}$$

At $v = 1$ the quotient is assigned its limiting value $u^2/2$. The same estimates as in Proposition 2.1 justify parameter differentiation and give

$$\mathfrak{S}_d(0) = 0, \qquad \mathfrak{S}_d'(u) = \Lambda_d(u). \tag{6.12}$$

Indeed the quotient equals $\int_0^u w v^w\, dw$ on any path in the half-plane. This identity integrates the completed spectral germ, so its pole subtraction and integration constant are explicit.

For comparison, the ordinary Hurwitz Stieltjes constants are defined by $\zeta(1+u,a) = u^{-1} + \sum (-1)^j \gamma_j(a) u^j/j!$, with $\operatorname{Re} a > 0$. Their classical parameter calculus follows at once from $\partial_a \zeta(s,a) = -s\zeta(s+1,a)$ [10]:

$$\partial_a \gamma_0(a) = -\zeta(2,a), \tag{6.13}$$

$$\partial_a \gamma_j(a) = (-1)^{j+1}\{\partial_s^j \zeta(2,a) + j\partial_s^{j-1}\zeta(2,a)\}, \qquad j \geq 1, \tag{6.14}$$

$$\int \gamma_j(a)\, da = \frac{(-1)^{j+1}}{j+1} \partial_s^{j+1}\zeta(0,a) + \text{constant}. \tag{6.15}$$

For the last identity expand $\partial_a \zeta(u,a) = -u\zeta(1+u,a)$ at $u = 0$. These familiar formulas provide a sign and factorial control when translating between ordinary Hurwitz constants and the strict-depth constants used in the new finite-part identities.

7 Finite reconstruction of every two-tail moment

The two-tail report proves a finite formula and a fixed coordinate space for every centered moment, but asks whether a uniformly specified finite prefix of the moments generates that space. We answer this question with an explicit bound and a rational reconstruction matrix. For equal tail orders the first q moments suffice, with an exact determinant. For unequal orders the bound below is deliberately not asserted to be optimal. All ranks in this section are ranks of matrices over $\mathbb{Q}$; the argument makes no assertion of arithmetic independence of zeta values.

The exact target is Question 8, "Finite generation by initial moments", in *Quartic Digamma Bridges, Harmonic Tail Moments, and Complex Resolvent Powers*. Its source theorem is "All even orders and all centered moments", with source label `tail:all-moments`; the inverse transform has source label `tail:inverse-transform`. The archive's Git blob is recorded in the provenance manifest; it was inspected at the repository commit printed on the title page. The source formulas are reproduced below to fix every normalization; the new argument begins with the finite-prefix theorem.

Let $2 \le p \le q$ be even, $b = q/2$, and

$$A_p(n) = \zeta(p) - H_n^{(p)}, \qquad D_r = D_{p,q}(-2r), \qquad D_{p,q}(s) = \sum_{n\ge 0} A_p(n)A_q(n)(n+1/2)^{-s}.$$

Outside its initial convergence half-plane, $D_{p,q}$ denotes its meromorphic continuation. Every displayed D_r is regular. The source two-tail theorem gives blocks $C_j = C_{p,q}(2j+1)$ as follows. Put $W = p + q$ and use the decreasing-index convention

$$\zeta(a,b) = \sum_{n>k\ge 1} n^{-a}k^{-b}, \qquad a \ge 2, \quad b \ge 1.$$

For a positive odd integer d, let

$$U_{p,q}(d) = \begin{cases} \zeta(p, q-d) + \frac{1}{2}\zeta(W-d), & d < q, \\ \displaystyle\sum_{h=1}^{\min\{(d-q+1)/2,\,p/2-1\}} \frac{\binom{d-q+1}{2h}}{d-q+1} B_{d-q+1-2h}\zeta(p-2h), & d > q, \end{cases} \tag{7.1}$$

$$R_{p,q}(d) = \begin{cases} 0, & d < W-1, \\ -\dfrac{B_{d-W+1}}{2}\left\{\dfrac{1}{p}\binom{d-q}{p-1} + \dfrac{1}{q}\binom{d-p}{q-1}\right\}, & d \ge W-1, \end{cases} \tag{7.2}$$

$$C_j = U_{p,q}(2j+1) + U_{q,p}(2j+1) + R_{p,q}(2j+1). \tag{7.3}$$

Every sum is finite; the two alternatives in (7.1) exhaust the possibilities because q is even. These blocks satisfy

$$D_r = \sum_{j=0}^{r} T_{rj}C_j, \qquad T_{rj} = \frac{\binom{2r}{2j}}{2j+1}B_{2r-2j}(1/2), \tag{7.4}$$

$$C_j = \sum_{r=0}^{j} B_{jr}D_r, \qquad B_{jr} = \binom{2j+1}{2r}2^{-2(j-r)}. \tag{7.5}$$

The B notation here denotes this finite matrix only when it has two indices; $B_n(x)$ and B_n are Bernoulli polynomials and numbers.

Why the inherited finite formula applies. For completeness, its finite-sum mechanism is short. The centered Hurwitz expansion of $A_p(n)$ contains only the powers $(n+1/2)^{1-p-2h}$. Hence A_pA_q contains only even inverse powers; asymptotic subtraction proves regularity of $D_{p,q}$

at $-2r$. All removed nonnegative even powers have continued sum zero, since $\zeta(-2h, 1/2) = 0$. Let CT_N mean the constant coefficient at positive infinity and put

$$S_r(N) = \sum_{n=0}^{N-1} (n + 1/2)^{2r} = \sum_{j=0}^{r} T_{rj} N^{2j+1}.$$

This odd polynomial has no constant term. Finite summation by parts, using $A_p(k-1) = A_p(k) + k^{-p}$, gives

$$D_r = \sum_{j=0}^{r} T_{rj} \, \mathrm{CT}_N \, E_{p,q,2j+1}(N),$$

where

$$E_{p,q,d}(N) = N^d A_p(N) A_q(N) + \sum_{k=1}^{N} k^{d-q} A_p(k) + \sum_{k=1}^{N} k^{d-p} A_q(k) + \sum_{k=1}^{N} k^{d-W}.$$

If $d < q$, the first ordered sum converges to $\zeta(p, q-d)$. If $d > q$, write $m = d - q$ and $F_m(N) = (B_{m+1}(N+1) - B_{m+1})/(m+1)$. The exact identity

$$\sum_{k=1}^{N} k^m A_p(k) = A_p(N) F_m(N) + \sum_{\ell=1}^{N} \frac{F_m(\ell - 1)}{\ell^p}$$

reduces its constant term to the Bernoulli sum in (7.1). The half-leading Faulhaber terms leave precisely one half of $\zeta(W-d)$ for each convergent ordered block. At $d = W - 1$ their two $-\gamma/2$ constants cancel the final power sum's $+\gamma$. The remaining three boundary products give exactly (7.2), by the uncentered Hurwitz expansion. This proves $\mathrm{CT}_N \, E_{p,q,d} = C_{p,q}(d)$ and explains the fixed finite-part normalization in (7.4). Finally the ordinary Bernoulli generating identity proves (7.5). These are inherited identities; no new priority is claimed for their derivation.

The source coordinate list is

$$C_0, \ldots, C_{b-1}, \quad Z_0, \ldots, Z_{b-1}, \qquad Z_k = \zeta(2k), \quad Z_0 = -\frac{1}{2}. \tag{7.6}$$

It has q specified entries. Using Z_0 instead of 1 is an invertible rational change of coordinates. For $j \geq b$ the source block formula is equivalently

$$C_j = \sum_{k=0}^{b-1} a_{jk} Z_k, \tag{7.7}$$

where, with $d = 2j + 1$ and $e = d - p - q + 2k + 1$,

$$a_{jk} = \begin{cases} B_e \left\{ \dfrac{\binom{d-q+1}{p-2k}}{d-q+1} + \dfrac{\binom{d-p+1}{q-2k}}{d-p+1} \right\}, & e \geq 0, \\ 0, & e < 0. \end{cases} \tag{7.8}$$

In this formula a binomial term is omitted unless its lower index is at least 2 and no greater than its upper index. This qualification matters: including a spurious lower index 0 changes the formula. The rational block is exactly the $k = 0$ term, since $Z_0 = -1/2$ and $\binom{u}{v}/u = \binom{u-1}{v-1}/v$.

Theorem 7.1 (A prescribed finite prefix)**.** *If $p < q$, put*

$$N = \frac{q(q-p)}{2} + \frac{q}{2} + p - 1. \tag{7.9}$$

The matrix

$$M = (a_{jk})_{b \leq j \leq N,\ 0 \leq k < b}$$

has column rank b over $\mathbb{Q}$. Consequently

$$\operatorname{span}_{\mathbb{Q}}\{D_0, D_1, \ldots\} = \operatorname{span}_{\mathbb{Q}}\{D_0, \ldots, D_N\} = \operatorname{span}_{\mathbb{Q}}\{C_0, \ldots, C_{b-1}, Z_0, \ldots, Z_{b-1}\}. \tag{7.10}$$

These are equalities of spans of actual numbers, regardless of any linear dependence among those numbers. The coefficient system on the q formal labels in (7.6) has rank exactly q. For $p = q$, all conclusions hold with $N = q - 1$.

The pair-dependent bound is at most $q(q-1)/2 + 1$, so the first $q(q-1)/2 + 2$ moments always suffice. The sharper diagonal assertion is proved separately below. The existence theorem answers the source finite-prefix question; it does not establish that the first q moments always suffice when $p < q$.

7.1 A differential description of the coefficient columns

Write $h(t) = \coth(t/2)$ and $m = q - p$. The source generator, or a direct termwise beta integration of the Bernoulli series, gives column functions

$$\begin{aligned} f_k(t) = {} & \frac{1}{2(p-2k)!(q-1)!} \int_0^t u^{p-2k}(t-u)^{q-1} h(u)\, du \\ & + \frac{1}{2(q-2k)!(p-1)!} \int_0^t u^{q-2k}(t-u)^{p-1} h(u)\, du. \end{aligned} \tag{7.11}$$

The first summand is omitted if $p - 2k < 2$. Both integrands otherwise have a removable endpoint singularity. Their Taylor series converge for $|t| < 2\pi$. The coefficient of $t^{2j+1}/(2j+1)!$ in f_k is a_{jk} for every $j \geq b$; lower coefficients will not be used. This follows from $h(u) = 2\sum_{n\geq 0} B_{2n} u^{2n-1}/(2n)!$ and Euler's beta integral.

For a vector $v = (v_0, \ldots, v_{b-1})$ over $\mathbb{C}$, put

$$f_v = \sum_{k=0}^{b-1} v_k f_k, \qquad Q(t) = \sum_{k=0}^{b-1} v_k \frac{t^{q-2k}}{(q-2k)!}.$$

Thus Q is an arbitrary even polynomial of degree at most q with $Q(0) = 0$. Differentiating the convolution integrals q times gives

$$2f_v^{(q)}(t) = \Big(Q^{(m)}(t) - Q^{(m)}(0)\Big)\, h(t) + (Q(t)h(t))^{(m)} =: \mathcal{L}_Q(t). \tag{7.12}$$

The subtraction $Q^{(m)}(0)$ enforces the omitted zero lower index in (7.8). Each side is an odd function holomorphic at zero.

Lemma 7.2 (A finite jet forces the polynomial to vanish)**.** *Suppose $m = q - p > 0$ and*

$$\operatorname{ord}_{t=0} \mathcal{L}_Q \geq 2N + 3 - q$$

with N as in (7.9). Then $Q = 0$.

Proof. Set $y = e^t$. For $r \geq 1$, direct differentiation shows that $h^{(r)}(t)$ has the form $P_r(y)/(y-1)^{r+1}$, where P_r is a polynomial of degree at most r, divisible by y. Indeed it is true for $h' = -2y/(y-1)^2$, and application of $y\, d/dy$ preserves the asserted form. Also $h = (y+1)/(y-1)$.

It follows by the product rule that

$$E(t) = (e^t - 1)^{m+1} \mathcal{L}_Q(t) = \sum_{\nu=0}^{m+1} P_\nu^*(t) e^{\nu t}$$

is an entire exponential polynomial. For $\nu = 0$ and $\nu = m+1$, $\deg P_\nu^* \leq p$; for $1 \leq \nu \leq m$, the degrees are at most q. To verify the endpoint degree statement, all terms involving $h^{(r)}$ with

$r \geq 1$ have an e^t factor and have exponential degree at most m, after the denominator is cleared. Only the h term reaches exponent 0 or $m+1$, and its polynomial factor is $2Q^{(m)} - Q^{(m)}(0)$, of degree at most p.

Therefore E is annihilated by the monic constant-coefficient differential operator

$$\mathcal{P}(\partial_t) = \partial_t^{p+1} \left(\prod_{\nu=1}^{m} (\partial_t - \nu)^{q+1} \right) (\partial_t - (m+1))^{p+1}. \tag{7.13}$$

Its order is $H = 2(p+1) + m(q+1)$. A holomorphic solution of this equation whose first H Taylor coefficients vanish is identically zero: the monic equation recursively expresses each higher derivative in the previous H derivatives. On the other hand, the hypothesis gives

$$\operatorname{ord}_0 E \geq (2N + 3 - q) + (m+1) = 2N + 4 - p = H.$$

Hence $E = 0$ and $\mathcal{L}_Q = 0$ meromorphically.

At every nonzero $\tau = 2\pi i\ell$, $h(t)$ has residue 2. Since the even integer m is at least 2, the pole of order $m+1$ in $(Qh)^{(m)}$ has coefficient $2m!Q(\tau)$. The other term of $\mathcal{L}_Q$ has at most a simple pole. Thus $Q(\tau) = 0$ for all nonzero integers ℓ. A polynomial with infinitely many zeros is zero. This proves the lemma. □

Proof of Theorem 7.1 for unequal orders. If $Mv = 0$, the odd Taylor coefficients of f_v at degrees $q+1, q+3, \ldots, 2N+1$ vanish. Differentiation q times kills every lower coefficient; the next possible degree is $2N + 3 - q$. Equation (7.12) and Lemma 7.2 give $Q = 0$, hence $v = 0$. The matrix has rank b already over $\mathbb{C}$, and therefore over $\mathbb{Q}$.

Its finite left inverse recovers all Z_k from $C_b, \ldots, C_N$. Equations (7.4)–(7.5) recover these blocks and the initial $C_0, \ldots, C_{b-1}$ from $D_0, \ldots, D_N$. All later blocks are given by (7.7); this proves all the span equalities. The early b coefficient rows are the identity on the $C_0, \ldots, C_{b-1}$ labels, and the later rows have rank b on the remaining labels. Thus the full formal rank is $2b = q$. □

7.2 An explicit rational reconstruction matrix

The preceding proof is effective without a search for a nonsingular minor. Define

$$L = (M^{\mathsf{T}}M)^{-1}M^{\mathsf{T}} = \frac{\operatorname{adj}(M^{\mathsf{T}}M)M^{\mathsf{T}}}{\det(M^{\mathsf{T}}M)}. \tag{7.14}$$

Every entry is an explicitly specified rational number obtained from the finite Bernoulli formula. The denominator is nonzero because M has full column rank; equivalently its real Gram matrix is positive definite. If $V = (C_b, \ldots, C_N)^{\mathsf{T}}$, then

$$(Z_0, \ldots, Z_{b-1})^{\mathsf{T}} = LV. \tag{7.15}$$

Together with (7.5), this is an explicit rational formula in the required prefix of actual moments. For any desired later D_r, insert (7.15) into (7.7) and then (7.4). This is a terminating exact algorithm, including when the actual coordinate values satisfy unknown rational relations.

One can also extract exactly q specified moments from this bounded prefix. Form their formal coefficient rows $W_r \in \mathbb{Q}^q$ by

$$(W_r)_j = T_{rj} \quad (0 \leq j < b), \qquad (W_r)_{b+k} = \sum_{j=b}^{r} T_{rj}a_{jk} \quad (0 \leq k < b),$$

where a term with $j > r$ is zero. Scan $r = 0, 1, \ldots, N$, retaining a row exactly when it increases rational rank. The theorem proves that this deterministic finite procedure retains q rows, with indices $r_1 < \cdots < r_q \leq N$. If A_* is the matrix of those rows, then

$$D_r = W_r A_*^{-1}(D_{r_1}, \ldots, D_{r_q})^{\mathsf{T}}$$

for every $r \geq 0$. This is an explicit basis-selection algorithm for the formal system, not a claim that its q actual numbers form an arithmetically independent set.

7.3 Equal orders: an exact determinant and a shorter prefix

If $p = q = 2b$, take the square late matrix with $b \le j \le 2b-1$. Write $j = b + t$ with $0 \le t < b$. Formula (7.8) is zero whenever $t + k < b - 1$, and on the antidiagonal $t + k = b - 1$ it equals $1/(t+1)$. Therefore

$$\det M = \frac{(-1)^{b(b-1)/2}}{b!} \neq 0. \tag{7.16}$$

This proves the asserted $N = q - 1$. It also gives a particularly simple reconstruction, for $t = 0, \dots, b-1$:

$$Z_{b-1-t} = (t+1)\left(C_{b+t} - \sum_{k=b-t}^{b-1} a_{b+t,k} Z_k\right). \tag{7.17}$$

The sum is empty at $t = 0$. In the coordinate ordering (7.6), the determinant of the complete first-q moment matrix is

$$\frac{(-1)^{b(b-1)/2}}{b!\,(2q-1)!!}. \tag{7.18}$$

Here the extra denominator is the product of the diagonal entries $1/(2r+1)$ of the forward Bernoulli transform.

7.4 A new concrete mixed example: orders two and six

For $(p, q) = (2, 6)$, the general theorem supplies a longer guaranteed prefix, but direct exact elimination proves that the first six moments already suffice. The last three block rows are

$$\begin{pmatrix} C_3 \\ C_4 \\ C_5 \end{pmatrix} = \begin{pmatrix} 2/3 & 5/12 & -1/12 \\ 5/6 & -7/24 & 1/12 \\ -47/60 & 1/2 & -3/20 \end{pmatrix} \begin{pmatrix} -1/2 \\ \zeta(2) \\ \zeta(4) \end{pmatrix},$$

whose determinant is $61/5760$. Its inverse is

$$\frac{1}{61}\begin{pmatrix} 12 & 120 & 60 \\ 344 & -952 & -720 \\ 1084 & -3800 & -3120 \end{pmatrix}. \tag{7.19}$$

For example, the rational row becomes the exact identity

$$61D_0 + 8204D_1 + 116000D_2 + 370048D_3 + 303360D_4 + 56320D_5 = -\frac{7808}{3}. \tag{7.20}$$

It has the ordinary absolutely convergent interpretation

$$\begin{aligned} &\sum_{n \ge 0} \left\{ P(x_n)A_2(n)A_6(n) - 11264x_n^4 - \frac{136960}{3}x_n^2 - \frac{88064}{5} \right\} = -\frac{7808}{3}, \\ &P(x) = 61 + 8204x^2 + 116000x^4 + 370048x^6 + 303360x^8 + 56320x^{10}, \qquad x_n = n + \frac{1}{2}. \end{aligned} \tag{7.21}$$

Indeed $A_2(n)A_6(n) = \frac{1}{5}x_n^{-6} - \frac{4}{15}x_n^{-8} + \frac{87}{200}x_n^{-10} + O(x_n^{-12})$, so the displayed bracket is $O(n^{-2})$. Also $A_2(n)A_6(n) = \psi'(n+1)\psi^{(5)}(n+1)/120$, giving a polygamma identity with exactly the same normalization.

What is still open. The first-q square coefficient matrix is nonsingular in every unequal pair checked by exact rational arithmetic through $q = 24$. This is finite evidence, not a proof for all p, q. The general theorem gives an explicit finite prefix and an exact rational inverse without assuming that stronger assertion. It also leaves untouched the arithmetic dimension of the actual number space: formal rank q is compatible with unknown rational relations among its zeta coordinates.

8 Prior art, coefficient conventions, and the source audit

8.1 Established ingredients and the attribution boundary

The normalized Gamma generator is closely connected with the classical Roman harmonic numbers. In the notation of Sesma [7, Proposition 3(a), equation (49)],

$$\frac{n!}{(1-t)_n} = \sum_{m\geq 0} c_n^{(m)} t^m = \frac{\Gamma(1-t)\Gamma(n+1)}{\Gamma(n+1-t)}. \tag{8.1}$$

Consequently $\mathcal{C}_m(n+1)/m! = c_n^{(m)}$ is the complete harmonic star sum with m unit indices. The complete harmonic-polynomial formula and the beta-logarithmic integral are also recorded in [7, equations (29)–(30)]; the relation of negative Roman degrees to strict harmonic sums appears in equations (24) and (26). The meromorphic functions $\mathcal{C}_m(x)$ used here are therefore natural continuations of established harmonic Bell polynomials. Their generator and their positive-integer specialization should not be presented as newly discovered objects.

The first ordinary-integral case is likewise known. Coffey [8, Proposition 3(b), equation (1.19)] gives

$$\gamma_1 = \zeta(2) + \int_0^1 \left(\gamma\psi(x) + \frac{\psi(x)^2 - \psi'(x)}{2}\right) dx. \tag{8.2}$$

Since $\mathcal{C}_2 = (\psi+\gamma)^2 + \zeta(2) - \psi'$, this is exactly

$$\int_0^1 \mathcal{C}_2(x)\, dx = \gamma^2 - \zeta(2) + 2\gamma_1 = 2(c_2 - \lambda_1), \tag{8.3}$$

where $c_k = [u^k]\Gamma(1+u)^{-1}$ and $\lambda_1 = [u]\zeta(1+u) = -\gamma_1$. Coffey's Appendix B also develops complex-degree Gamma derivatives and their Bell-polynomial and polygamma descriptions.

Young's work on height-one multiple zeta functions establishes the reciprocal-Gamma singular Laurent data at $s = 1$ and expressions for the first regular coefficient in terms of Bernoulli numbers of the second kind and multiple Hurwitz integrals [4, 5, 6]. It also relates strict-depth Laurent constants to Ramanujan sums of complete harmonic star sums. In particular, the Cauchy-number series for the linear coefficient in this article is existing literature in equivalent notation. To make the equivalence explicit, put

$$\lambda_d = [u]\zeta(1+u, \{1\}^{d-1}), \qquad a_n = \frac{1}{(n-1)!}\int_0^1 (r)_n\, dr. \tag{8.4}$$

If $t/\log(1+t) = \sum_{n\geq 0} b_n t^n$, then

$$\frac{a_n}{n} = \sum_{k>n} |b_k|, \qquad \sum_{n\geq 1} \frac{a_n}{n^{d+2}} = \sum_{k\geq 1} |b_k| H_{k-1}^{(d+1)}. \tag{8.5}$$

The first equality follows by comparing coefficients in

$$\sum_{n\geq 0} \frac{z^n}{n!}\int_0^1 (r)_n\, dr = \frac{-z}{(1-z)\log(1-z)} = \frac{1 - \sum_{k\geq 1}|b_k| z^k}{1-z}. \tag{8.6}$$

Here $\sum_{k\geq 1}|b_k| = 1$, obtained by letting $z \uparrow 1$ in $-z/\log(1-z) = 1 - \sum_{k\geq 1}|b_k| z^k$. The second equality in (8.5) then follows by Tonelli's theorem. Thus

$$\lambda_d = c_{d+1} + \sum_{n\geq 1} \frac{a_n}{n^{d+2}} \tag{8.7}$$

is precisely Young's coefficient formula after a change of Cauchy-number convention, rather than an independent new evaluation.

The contribution developed here is the explicit residue organization needed for the finite-part problem: the scalar Gamma–Appell normalizer, its uniqueness up to constants, its finite decaying defect, and its comparison with the exact differential normalizer. The ordinary integral represents the established strict-depth coefficient in terms of Gamma and polygamma functions. These statements are proved directly in the article. The source search supplies attribution for their ingredients; it does not establish worldwide priority for every possible reformulation.

8.2 A sign description in Young's 2024 paper

There is a specific convention issue worth correcting before the formulas are imported into a symbolic library. In the standard Stieltjes convention,

$$\zeta(1+u) = \frac{1}{u} + \gamma - \gamma_1 u + O(u^2). \tag{8.8}$$

Thus the literal linear coefficient is $-\gamma_1$. Young's earlier lecture explicitly defines the regular coefficients with the factors $(-1)^i/i!$ and, on its numbered slides 3 and 6, correctly identifies $-\gamma_1^{[d]}(1)$ as the linear coefficient [6].

In [5, equations (4.1)–(4.2) and Corollary 3], the verbal description identifies the same signed Stieltjes constant with the literal linear coefficient. With the explicit notation (8.4), the unambiguous translation of equation (4.2) is

$$\nu_{0,d} - \nu_{-1,d-1} = c_{d+1} - \lambda_d, \tag{8.9}$$

where the two series in the source's notation are

$$\nu_{0,d} = \sum_{n\geq 2} \frac{(-1)^n}{n} \zeta(n, \{1\}^d), \qquad \nu_{-1,d-1} = \sum_{n\geq 2} \frac{(-1)^n}{n-1} \zeta(n, \{1\}^{d-1}). \tag{8.10}$$

The second series is interpreted as its ordinary convergent alternating sum when $d = 1$; the other cases here are absolutely convergent. At depth one,

$$\nu_{0,1} - \nu_{-1,0} - c_2 = \gamma_1 = -\lambda_1, \tag{8.11}$$

which exhibits the sign directly. The appropriate correction is to describe $\gamma_1^{[d]}(1)$ as the *negative* of the linear coefficient, or to translate every formula consistently into the literal-coefficient notation λ_d. The equations for the conventional signed constants need not be reversed.

The finite-part convention is a separate issue. In the reviewed quartic-tail and Barnes incoming articles, the quartic constants denoted θ and β_1 represent the same strict coefficient $[u]\zeta(1+u,1,1)$, and the finite parts use a cutoff in x itself. No additional incompatibility between those conventions was established in the audit. A change of cutoff coordinate, however, must be transported explicitly before two finite-part formulas are compared; a change of Laurent-coefficient sign must not be used to absorb such a coordinate change.

9 Further research questions and concrete next targets

The following programme emphasizes exact identities and structural reduction. Its first two items sharpen source questions partially answered here; the subsequent items follow from the proved formulas. Numerical experiments may guide these questions, but their desired outputs are identities, rank theorems, and analytic proofs.

9.1 The first q moments for unequal even orders

Conjecture 9.1 (The optimal consecutive prefix)**.** *For all even $2 \le p < q$, the square matrix*

$$(a_{jk})_{q/2\le j\le q-1,\ 0\le k<q/2}$$

is nonsingular over $\mathbb{Q}$. Equivalently, the first q centered moments have the full formal rank q.

The article proves a larger finite prefix for every pair and the conjectured prefix on the diagonal. Exact calculations verify the unequal assertion through $q = 24$, but this is a finite test. A useful next proof would replace the general exponential-polynomial zero count by a sharper property of the restricted even-polynomial space in (7.12). An explicit determinant formula would also explain the nontrivial rational numerators already visible for $(2, 4)$ and $(2, 6)$. A sign pattern alone would be insufficient unless it proves nonvanishing for every order.

9.2 Can the mixed fifth coordinate be eliminated?

Theorem 5.1 singles out the holomorphic quantity $J'(1)$. The first unresolved question is whether it has an exact reduction to the height-one coefficients and ordinary zeta/Stieltjes constants already present in that theorem. It is not appropriate to declare a new independent constant merely because it appears in one reduction. Conversely, a small numerical integer relation would not establish its reducibility.

A more algebraic version asks for a scalar polynomial in ψ whose coordinate finite part contains only a prescribed highest-depth coefficient and ordinary constants. The scalar Gamma–Appell theorem controls the nondecaying residue polynomial at infinity. Its explicit defects, starting at degree four, show why this statement alone does not settle all convergent lower-depth contributions. A promising next target is to classify the residue defects modulo discrete differences whose logarithmic sums reduce by finite harmonic algebra.

9.3 An all-degree classification of the mixed coordinates

The differential sequence $\mathcal{C}_m$ has exactly one height-one residue word. The raw scalar power ψ^m introduces products involving higher polygammas. At degree five the additional coordinate is represented by FP $\int \psi(\psi')^2$. At degree six and beyond, one can ask: which differential monomials survive integration by parts after every finite endpoint constant is retained?

A precise project is to grade monomials by $\mathrm{wt}(\psi^{(r)}) = r+1$, quotient them by total derivatives, and compute the finite-dimensional residue-coordinate image at each weight. This must keep endpoint evaluation as part of the map: discarding a total derivative without its finite constant would produce a different problem. The desired result is a recursion for the surviving spectral words, with a proved coordinate count, rather than a list of fitted numerical bases.

9.4 Shifted and rational-cutoff Gamma normalizations

Replacing the base point 1 by a parameter a naturally produces $\Gamma(a-t)\Gamma(x)/(\Gamma(a)\Gamma(x-t))$ and differences of Hurwitz polygammas. The special base point in this article makes the origin residue exactly linear in t, which is responsible for the ordinary integrals at every degree $m \ge 2$. For general a, a new subtraction is required.

Find the canonical joint subtraction in x and t, prove the resulting integral bridge, and express it in the ordered multiple-Hurwitz Laurent coordinates already developed in the incoming Barnes report. Rational a should connect the resulting formulas with congruence-restricted harmonic sums and cyclotomic polylogarithms. The question is the exact compatibility of the normalizations, not just formal substitution in an unshifted formula.

9.5 Four tails and higher products

For products $\prod_{i=1}^{k}(\zeta(p_i) - H_n^{(p_i)})$ with even p_i and even k, the centered even-power asymptotics persist. Finite summation by parts produces higher-depth convergent zeta coordinates. Determine a finite formal coordinate system and prove an effective initial-moment generation theorem for even $k \geq 4$. Odd numbers of tails instead produce odd centered powers and can have poles at the target spectral values; their finite-part moments constitute a separate extension. The two-tail proof suggests a route through differential operators on products of $\coth(t/2)$, but higher pole multiplicities and overlaps of coordinate columns must be handled explicitly. Merely observing stabilization in low-order matrices would not prove a uniform theorem.

9.6 Exact primitives at algebraic or rational arguments

The functions $\mathfrak{P}_m$ provide explicit normally convergent logarithmic primitives on a slit plane. Evaluate their values at rational arguments and their reflection or multiplication combinations. One may seek reductions to Barnes multiple-Gamma functions, multiple Hurwitz derivatives, or cyclotomic polylogarithms. The removable values (3.10) and the shift identity (6.10) supply exact boundary conditions. Any proposed primitive identity should match both its derivative and one normalization value; matching derivatives alone leaves an undetermined constant.

9.7 Higher Laurent jets and their functional relations

Equation (2.5) supplies every regular Laurent coefficient through a fixed logarithmic Mellin kernel. Determine which identities among those logarithmic moments arise from polylogarithmic functional equations, shuffle products, and the spectral primitive (6.11). A particularly concrete question is whether selected mixed derivatives in the sixth-power calculation can be expressed using only finitely many height-one jets plus a small collection of higher-depth germs.

9.8 Proof certificates and unresolved ladder identities

The finite harmonic identities in the fifth-power ledger are suitable for independent formal verification: only exact quasi-shuffle multiplication, the prefix rule, explicit low-weight reductions, and polynomial constant-term extraction are needed. The analytic layer can be separated into normal convergence, the contour remainder, and the Mellin continuation lemmas. This provides a manageable route to integrating proof certificates into ProveIt.

The manuscript's remaining S_6 and revised S_8 ladder candidates continue to deserve exact functional-equation research. Their existing high-precision certificates should guide the search for an identity or an obstruction, while remaining explicitly distinct from a proof of equality. No result of this article changes their status.

10 Verification, provenance, and integration

The accompanying archive is an editable research package. It contains this article in modular and single-file LaTeX, the compiled PDF, the exact calculation scripts, numerical evaluators, their recorded JSON outputs, a build script, a checksum manifest, and proposed repository integration notes. The scripts require Python, SymPy, and mpmath; no network access or private data is needed to replay them.

10.1 What the calculations verify

The fifth-power exact replay checks 87 identities using finite harmonic quasi-shuffle algebra. They include all principal coefficients, the constant Laurent coefficients at zero, both finite residue-sum identities, their asymptotic constants, the nonlogarithmic sums, the spectral substitution, and

every endpoint entry used in the integration by parts. These are symbolic finite identities for arbitrary cutoff N, together with explicitly stated classical multiple-zeta reductions; the computation does not infer the identities by sampling N.

The tail reconstruction replay checks 1,582 exact assertions. These include the forward and inverse Bernoulli transforms, the agreement of the compact coefficient matrix with the inherited finite formula, the diagonal determinant, the displayed rational inverse, the orders-$(2,6)$ identity, and finite instances of the exponential-polynomial degree counts. It also verifies the first-q square matrix for all 78 even pairs $2 \le p \le q \le 24$. Those finite checks support the stronger conjecture but are not its proof. The all-pairs finite-prefix theorem rests on the differential and pole argument in Section 7.

The independent Gamma-normalizer replay checks 202 exact assertions through degree eight and integer pole labels zero through eight, including the removable values and the first nonzero residues. Separate ordinary-integral computations for $m = 2, \dots, 7$ use analytic Gamma coefficients on $[0,1]$ and the completed logarithmic Mellin kernel on $(0,\infty)$. The largest recorded absolute discrepancy is below $7.403 \cdot 10^{-58}$ at 70 working decimal digits. A local Taylor switch is changed independently to test cancellation handling. The scalar normalization has its own 94-check finite polynomial replay, including the inverse-Gamma finite-part convolution through degree ten. Together the four exact suites verify 1,965 assertions.

Independent comparison	Finest recorded absolute discrepancy	Type
Fifth-power finite part	$< 3.83 \cdot 10^{-48}$	Taylor-subtracted quadrature versus harmonic arithmetic with asymptotic tail
Mixed integral I	$< 1.02 \cdot 10^{-48}$	The same two independent analytic representations
Orders-$(2,6)$ rational sum	$< 2.40 \cdot 10^{-62}$	Hurwitz-tail sum with separately derived centered tail
Gamma ordinary integrals	$< 7.403 \cdot 10^{-58}$	Bell/Gamma coefficients versus completed Mellin kernel

All entries in this table are floating-point diagnostics. They are not interval certificates and are not presented as proofs. In particular, the Gamma evaluator records a bound for its omitted analytic kernel tail, but that bound does not enclose quadrature, local Taylor truncation, or rounding error. The mathematical conclusions follow from the analytic and exact algebraic arguments in the text.

10.2 Source and editorial controls

The provenance manifest records the inspected repository commit, the source archive and relevant member hashes, and the external primary references. The main manuscript was checked for prior results before the new targets were selected. The audit does not purport to certify every page of the repository. In particular, the already proved Choi harmonic conjectures and the proved S_4 identity are not rediscovered here, and S_6 and the revised S_8 remain explicitly open.

The integration notes propose where to place the new sections and how to update the three source questions. They also provide the precise Stieltjes sign translation, so an importer can retain the literal-coefficient notation consistently. No repository file has been silently altered, and the package is intended for review and integration through the project's normal process.

The broader literature also contains harmonic-zeta Stieltjes families with different indexing conventions; see, for example, Kargın–Dil–Cenkci–Can [9]. All uses of such constants in future extensions should name the defining germ and its factorial and sign convention explicitly, as done here.

References

[1] V. Reshetnikov, *ProveIt*, GitHub repository, manuscript and incoming reports inspected at commit `b7667ae40a2957eed9a53d8559d53a3ee20dd1c1`, 11 October 2026. https://github.com/VladimirReshetnikov/ProveIt. Repository paths and file hashes are recorded in the accompanying `provenance/` directory.

[2] *Quartic Digamma Bridges, Harmonic Tail Moments, and Complex Resolvent Powers*, ProveIt incoming report, 11 October 2026. Archive `ProveIt_Quartic_Tails_Resolvents_2026-10-11.zip`, in `docs/incoming/` at the snapshot in [1]. Principal source sections: `mixed_even_tails.tex`, `mixed_even_tail_generators.tex`, and `research_questions.tex`.

[3] *Barnes Resolvents and Higher Harmonic Identities*, ProveIt incoming report, 11 October 2026, in the incoming directory at the snapshot in [1]. Used for the height-one Laurent normalization and the comparison of quartic finite-part conventions.

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[6] P. T. Young, *Global series for height 1 multiple zeta functions*, public lecture slides, 4 July 2023, 17 numbered slides with overlays. https://iecl.univ-lorraine.fr/wp-content/uploads/2023/07/tuesday-12-young-p.pdf. The Laurent convention occurs on numbered slides 3 and 6; complete harmonic star sums and their Ramanujan sums occur on slides 16 and 17.

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